



Lehrstuhl für Schaltungsentwurf

Technische Universität München

Prof. Dr.-Ing. Ralf Brederlow



Lab Report

Lab CMOS Voltage Regulation Circuit design

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May 5, 2024

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0.1 Task 1: Parameter Extraction

0.1.1 Extract the threshold voltage V_{T0} without considering the body effect (V_{SB})=0

NMOS

This is the waveform of NMOS $V_{gs}/\sqrt{I_d}$ in linear scale. We plot straight lines in the saturation region and find the intersection with the x-axis. As V_g increases, the MOS first enters the saturation region, then the linear region.

Table 1: NMOS V_{th}

5Wmin, 5Lmin	5Wmin, 2Lmin	5Wmin, Lmin	2Wmin, 5Lmin	2Wmin, 2Lmin	2Wmin, Lmin	Wmin, 5Lmin	Wmin, 2Lmin	Wmin, Lmin
0.3880	0.3709	0.3463	0.4023	0.3829	0.3550	0.4177	0.3952	0.3631

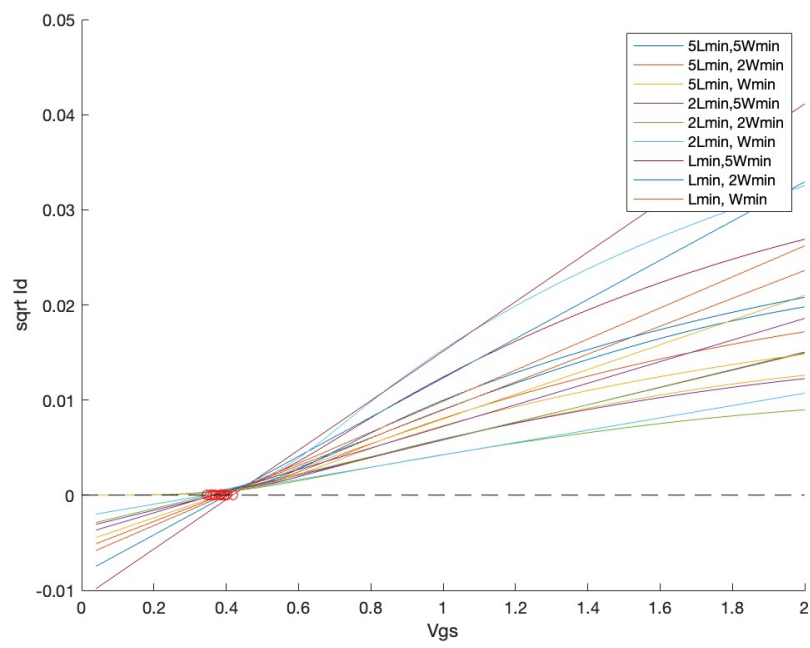


Figure 1: nmos, v_{th}

PMOS

PMOS: the same concept as NMOS. When we calculate V_{th} , we use the formula 2 – intersection_points because I set V_s to $2V$.

Table 2: PMOS V_{th}

5Wmin, 5Lmin	5Wmin, 2Lmin	5Wmin, Lmin	2Wmin, 5Lmin	2Wmin, 2Lmin	2Wmin, Lmin	Wmin, 5Lmin	Wmin, 2Lmin	Wmin, Lmin
0.3662	0.3583	0.3494	0.3878	0.3789	0.3682	0.4081	0.3983	0.3859

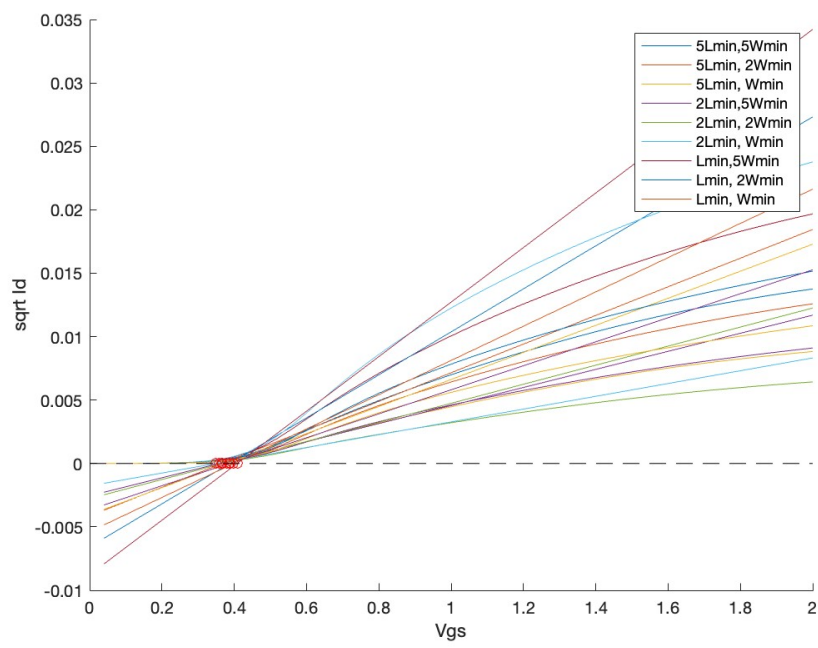


Figure 2: pmos, vth

0.1.2 Extract the parameter $K = \mu C_{ox}$

We have to take the points in the linear region of I_d .

NMOS

NMOS: According to the formula in the book, we calculate $K = \text{slope} \times \frac{L/W}{V_{ds}}$, because I set $V_{ds} = 0.9V$, so I calculate $\text{slope} \times \frac{L/W}{0.9}$.

Table 3: NMOS K

NMOS K								
5Wmin, 5Lmin	5Wmin, 2Lmin	5Wmin, Lmin	2Wmin, 5Lmin	2Wmin, 2Lmin	2Wmin, Lmin	Wmin, 5Lmin	Wmin, 2Lmin	Wmin, Lmin
0.3566×10^{-4}	0.3835×10^{-4}	0.4200×10^{-4}	0.2765×10^{-4}	0.2972×10^{-4}	0.3239×10^{-4}	0.2213×10^{-4}	0.2373×10^{-4}	0.2562×10^{-4}

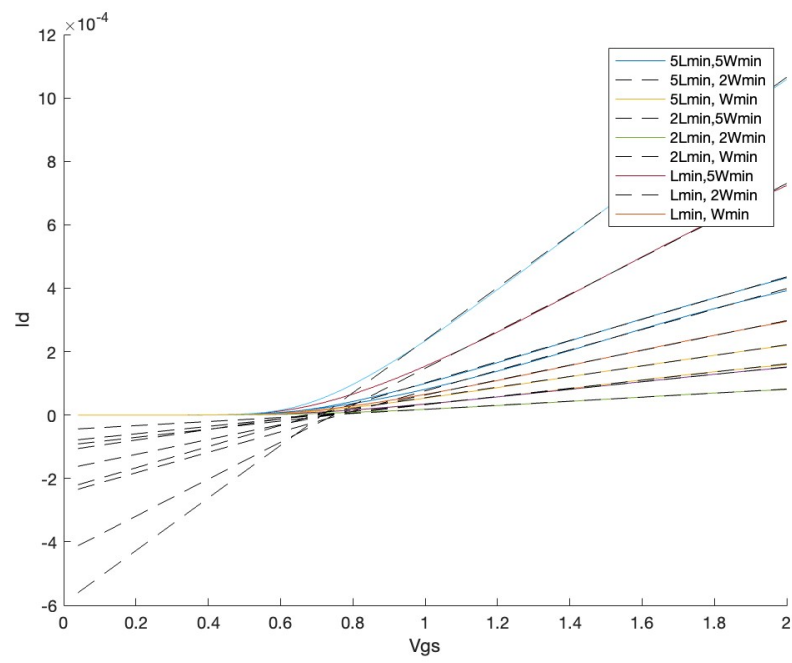


Figure 3: nmos, k

PMOS

PMOS: same concept as NMOS.

Table 4: PMOS K

PMOS K								
5Wmin, 5Lmin	5Wmin, 2Lmin	5Wmin, Lmin	2Wmin, 5Lmin	2Wmin, 2Lmin	2Wmin, Lmin	Wmin, 5Lmin	Wmin, 2Lmin	Wmin, Lmin
0.2804×10^{-4}	0.2803×10^{-4}	0.2738×10^{-4}	0.2074×10^{-4}	0.2073×10^{-4}	0.2029×10^{-4}	0.1477×10^{-4}	0.1483×10^{-4}	0.1472×10^{-4}

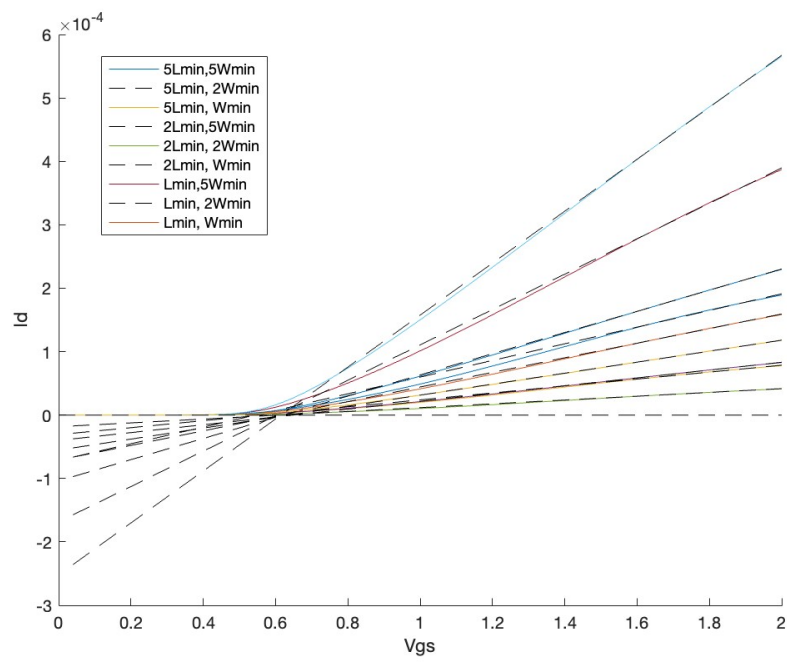


Figure 4: pmos, k

0.1.3 Extract the channel length modulation factor λ

NMOS

NMOS: first, we take the I_d , V_{ds} curve for NMOS, then measure the slope in the saturation region and measure the intersection with the y-axis. To get $\lambda = \text{slope}/\text{intersection}$.

Table 5: NMOS λ

5Wmin, 5Lmin	5Wmin, 2Lmin	5Wmin, Lmin	2Wmin, 5Lmin	2Wmin, 2Lmin	2Wmin, Lmin	Wmin, 5Lmin	Wmin, 2Lmin	Wmin, Lmin
0.0548	0.0538	0.0524	0.0756	0.0737	0.0711	0.0963	0.0927	0.0879

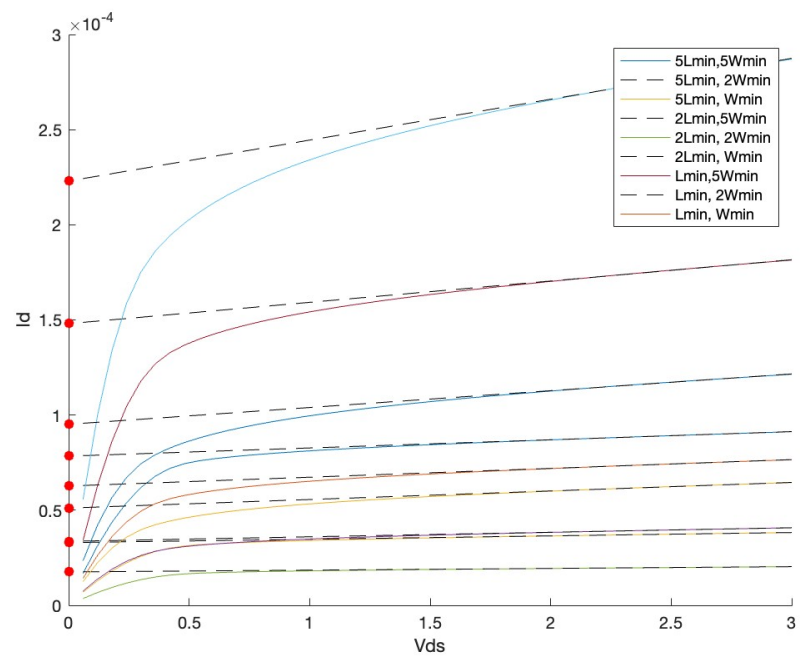


Figure 5: nmos, lambda

PMOS

Same concept as NMOS

Table 6: PMOS λ

5Wmin, 5Lmin	5Wmin, 2Lmin	5Wmin, Lmin	2Wmin, 5Lmin	2Wmin, 2Lmin	2Wmin, Lmin	Wmin, 5Lmin	Wmin, 2Lmin	Wmin, Lmin
0.0620	0.0611	0.0598	0.0695	0.0681	0.0657	0.0727	0.0703	0.0666

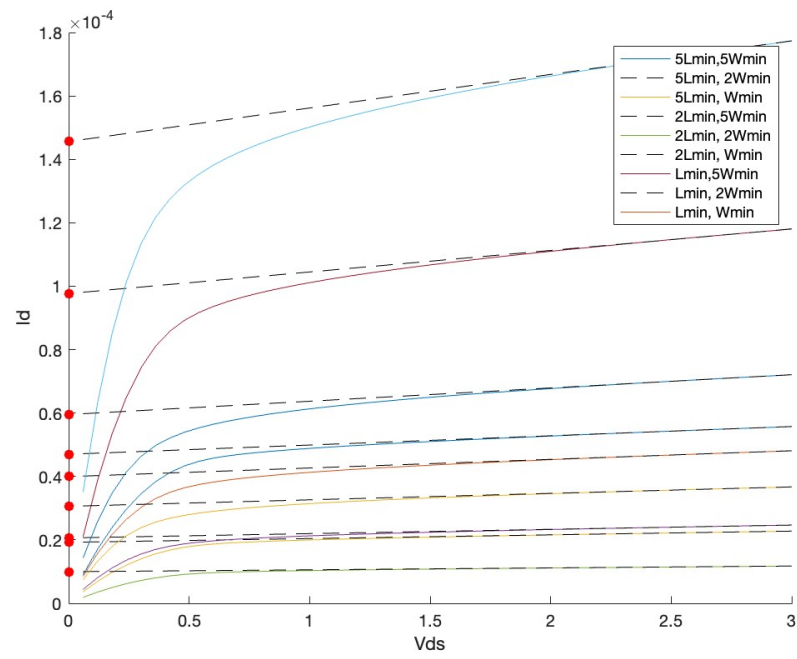


Figure 6: pmos, lambda

0.1.4 Compute the simulated transistor characteristics to your own model

Ideal model:

NMOS, V_{gs} , ideal

V_{ds} fixed at 0.9V. V_{gs} sweep from 0 to 2V.

PMOS, V_{gs} , ideal

V_{ds} fixed at 0.9V. V_{gs} sweep from 0 to 2V.

NMOS, V_{ds} , ideal

V_{gs} fixed at 0.9V. V_{ds} sweep from 0 to 3V.

PMOS, V_{ds} , ideal

V_{gs} fixed at 0.9V. V_{ds} sweep from 0 to 3V.

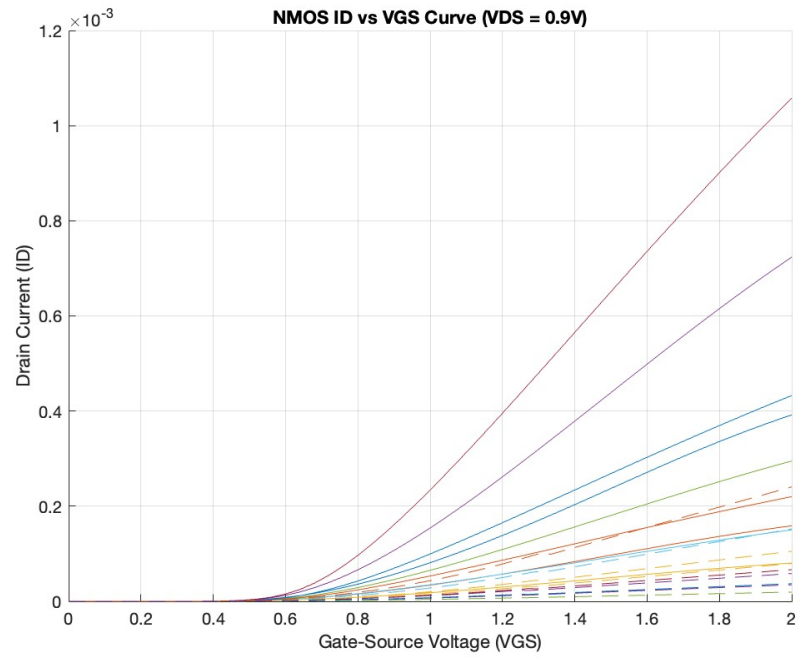


Figure 7: nmos, ideal vgs curve

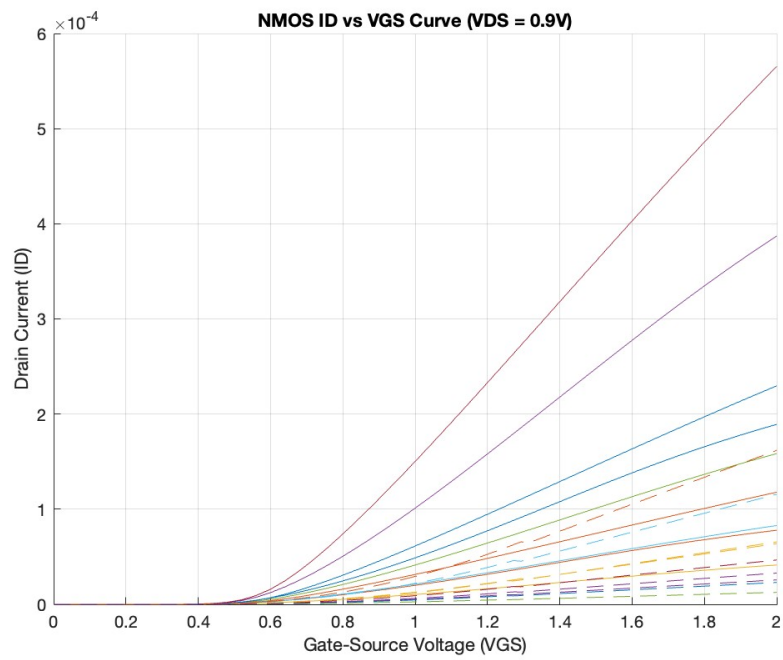


Figure 8: pmos, ideal vgs curve

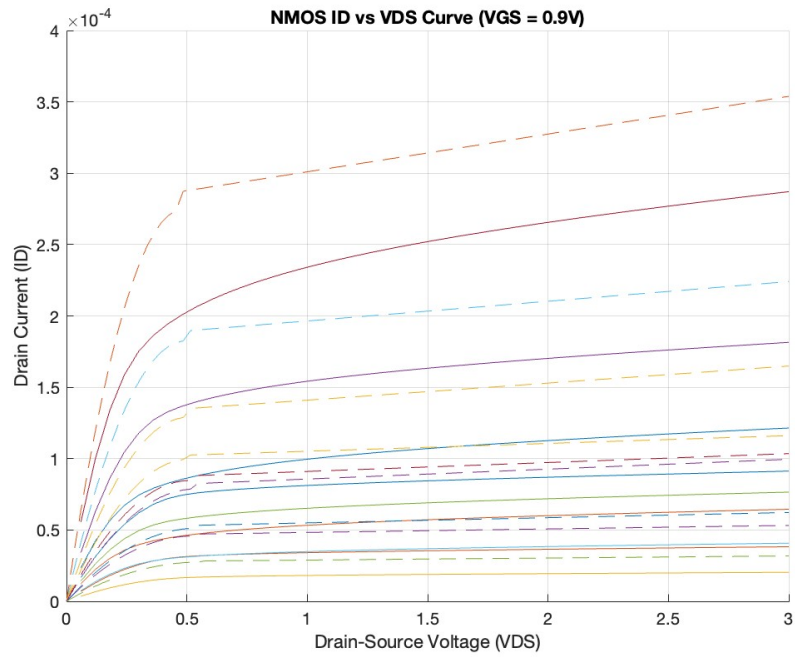


Figure 9: nmos, ideal vds curve

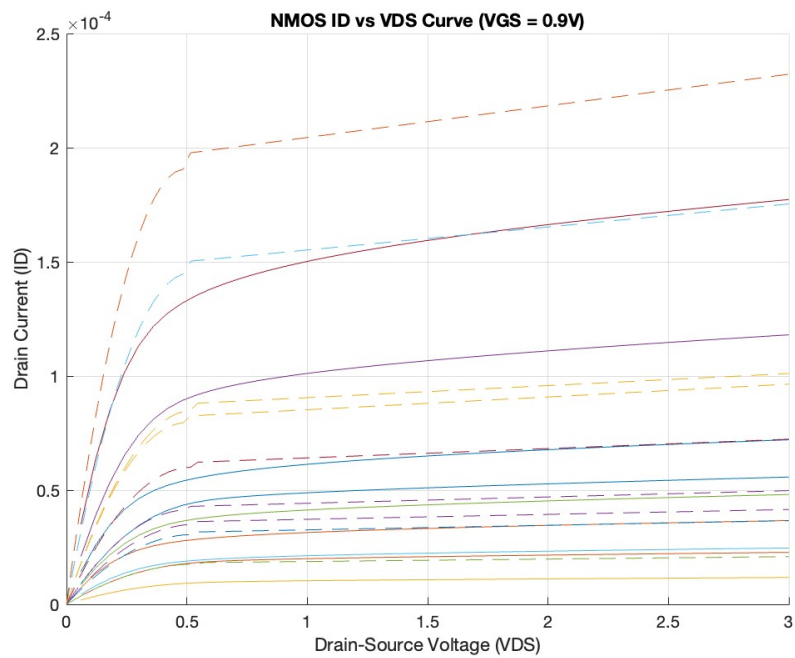


Figure 10: pmos, ideal vds curve

APPENDIX

A Schematic and Layout

- NMOS

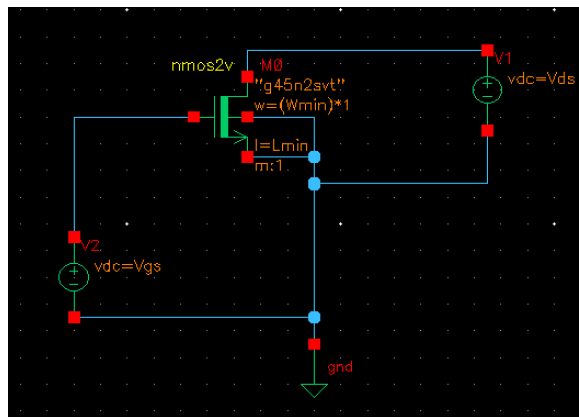


Figure A.1: NMOS Schematic

- PMOS

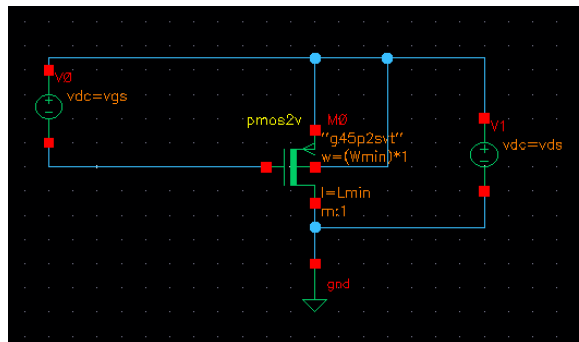


Figure A.2: PMOS Schematic

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1.1 Answer to the questions

Table 1.1: Specifications

Specification	Condition	Note
V_{DD}	1.5V–1.8V	Positive supply voltage
V_{SS}	0V	Negative supply voltage
V_{out}	$1.2V \pm 5\%$	Reference voltage
PSRR	$\geq 60\text{dB}$	Power supply rejection ratio
I_{load}	10 μ A–10mA	Current to be supplied
P	Minimize	$\leq 1\text{mW}$

1.1.1 Why are solutions based on resistors and diodes without feedback (e.g., a simple resistive divider) insufficient?

Figure 1.1 shows an example of an LDO without a feedback loop. In this configuration, the output voltage is regulated by the fixed voltage drop across a zener diode. However, this solution has limitations. When the load current (I_L) is too high, the voltage across the resistor R_v will also increase, potentially exceeding the operating limits of the zener diode and causing a drop in V_{out} . Including a feedback loop enables the regulator to react to variations in the output current, maintaining a stable output voltage.

1.1.2 Estimate the value for each of the performances in your circuit based on hand calculations. Use the first-order transistor model parameters extracted in Task 1.

First we need to discuss components used to build LDO circuit

1.1.2.1 LDO schematic discuss

In schematic 1, the simple LDO consists of few elements

- Error Amplifier: used to amplify our error and adjust the PMOS current
- PMOS pass gate
- Vref reference voltage: control our output voltage
- Voltage divider R1, R2: feedback voltage
- some parallel or series capacitance/resistance to improve stability

1.1.2.2 Why do we use PMOS instead of NMOS?

The power dissipation of a pass gate is given by $I_{\text{load}} \times V_{\text{dropout}}$, where I_{load} is the load current and V_{dropout} is the dropout voltage. PMOS structures often have lower dropout voltages than NMOS, resulting in lower power dissipation. However, NMOS transistors have inherently lower output resistance. Therefore, when low dropout voltage is not a primary concern, NMOS may be preferred to simplify regulator compensation.

1.1.2.3 Dropout voltage calculation

1.2

The dropout voltage (V_{dropout}) is calculated as follows:

$$V_{\text{dropout}} = V_{\text{DD}} - V_{\text{DS}_{\text{passgate}}}$$

We assume the passgate operates in the saturation region. From the formula

$$I_{\text{D}} = \frac{1}{2} \mu_{\text{p}} C_{\text{ox}} \frac{W}{L} (V_{\text{GS}} - V_{\text{th}})^2$$

For fixed V_{DS} , we can calculate $\frac{W}{L}$. We discuss two extreme cases where we have maximum load current and minimum load current. Also, V_{GPMOS} cannot exceed $V_{\text{DD}} - V_{\text{th,p}}$.

Table 1.2: PMOS Assumptions

Variable	Case 1	Case 2
R_{load}	0.12 k Ω	0.12 M Ω
$\mu_p C_{\text{ox}}$	30 $\mu\text{A}/\text{V}^2$	30 $\mu\text{A}/\text{V}^2$
V_{DS}	0.3 V	0.3 V
I_{D}	10 mA	10 μA
$\frac{W}{L}$	7400	7.4
V_{th}	0.4V	0.4V
V_{DD}	1.5V	1.5V
V_{GPMOS}	max 1.1 V	max 1.1 V

1.1.2.4 PSRR

The PSRR is calculated using the formula:

$$\text{PSRR} = 20 \log_{10} \left(\frac{V_{\text{in,RPL}}}{V_{\text{out,RPL}}} \right)$$

This paper demonstrates how to calculate[1] Assume the parameter due to this paper[2] Feedback can effectively reduce output resistance. The feedback impedance (Z_{fb}) is given by:

$$Z_{\text{fb}} = \frac{V_T}{I_T} = \frac{R_{\text{out}}}{1 + A \times \beta}$$

Where V_T is the thermal voltage, I_T is the bias current, R_{out} is the output resistance, A is the open-loop gain, and β is the feedback factor.1.3

The Power Supply Rejection Ratio (PSRR) can be calculated using the formula:

General Formula:

$$\text{PSRR} = 20 \log \left(\frac{Z_{\text{fb}} + r_{\text{ds}}}{Z_{\text{fb}}} \right) = 20 \log \left(1 + \frac{r_{\text{ds}}}{Z_{\text{fb}}} \right)$$

So if Z_{fb} is very small, we have good PSRR. To make it small, we need high gain. However, when gain drops at high voltage, we also experience higher noise.

PSRR at low frequency: In the low-frequency range, capacitors are viewed as open loops. Therefore, PSRR can be approximated as:

$$\text{PSRR} \approx 20 \log(A \times \beta)$$

Referring to Figure 1.6, $\text{PSRR} \approx 20 \log 6500 = 76\text{dB}$

In the middle frequency range ($< 100\text{kHz}$), we must consider the gain of the error amplifier. As the gain decreases, our PSRR will also decrease.

Therefore, PSRR can be approximated as:

PSRR(mid freq)

$$\text{PSRR} \approx 20 \log \left(1 + \frac{A}{1 + \frac{s}{P}} \times \beta \right)$$

where P is the corner frequency of error amplifier.

In high frequency: from the above equation we can calculate PSRR in high frequency is very small.

In ultra-high frequencies, the load capacitance (C_{load}) becomes dominant. As a result, PSRR can be approximated as:

$$\text{PSRR} \approx 20 \log (1 + s \times C_{\text{load}} \times r_{\text{ds}})$$

In ultra-high frequencies, the load capacitance (C_{load}) becomes negligible. As a result, PSRR stabilizes and can be approximated as:

PSRR(high freq)

$$\text{PSRR} \approx 20 \log \left(\frac{R_{\text{esr}} + r_{\text{ds}}}{R_{\text{esr}}} \right)$$

This picture shows PSRR in different frequency.1.4

This picture shows the matlab simulation of PSRR from low to middle frequency.1.5

My PSRR at low frequency is about 50 dB.

The equation for r_{ds} in the saturation region is:

$$r_{\text{ds}} = \frac{1}{\lambda I_{\text{D}}}$$

where λ is approximately 0.7 from Task 1. For calculation simplicity, we assume $\beta = 1$. In worst case r_{ds} should be largest, assume $1/(0.07 \times 10\text{m})$.

Table 1.3: Our assumptions

$R_{\text{out}_{EA}}$	300k Ohm
$A \times \beta = A$	1000(60dB)
r_{ds}	14285 Ohm
I_{D}	10mA

1.1.2.5 Open Loop Gain

Refer to Figure 1.2 and paper [2], we first decide our parameters. For simplicity, we assume $\beta = 1$.

Open loop gain $G_{op} = G_{pmos} \times G_{EA}$.

$$gm_{pmos} = \frac{2 \cdot I_D}{V_{ov}} = 0.1$$

$$G_{pmos} = gm_{pmos} \times R_{OPMOS} = 6.5.$$

Table 1.4: LDO parameter

R_{O_PMOS}	65 Ohm
G_{EA}	1000 (60 dB)
I_D	10 mA
$V_{ov} = V_{gs,pmos} - V_{th,p}$	0.2 V
R_{O_EA}	300k Ohm
R_{ESR}	2 Ohm
C_{load}	10 uF
C_{PMOS}	200pF

There will be three poles and one zero:

- pole1 = $\frac{1}{2\pi(R_{OPMOS} + R_{ESR})C_{load}} = 238Hz$
- pole2 = $\frac{1}{2\pi R_{ESR}C_{BP}} = 159kHz$
- pole3 = $\frac{1}{2\pi R_{OEA}C_{PMOS}} = 2.65kHz$
- zero1 = $\frac{1}{2\pi R_{ESR}C_{load}} = 7.96kHz$

When we plot bode plot, we ignore p2 because it's in high frequency.

In figure1.6, we can see that the dc gain is $6.5 \times 1000 = 76$ dB, and two poles and one zero.

Lastly, we should discuss the closed-loop response. (Assume $\beta = 1$). The equation is $V_{out} = V_{ref} \cdot \frac{V_{out}}{V_{ref}} + V_{DD} \cdot \frac{V_{out}}{V_{DD}}$. The second term is ignored because $\frac{V_{out}}{V_{DD}}$ is very small (assuming we have high PSRR).

$$\frac{V_{out}}{V_{ref}} = \frac{G_{op}}{1 + G_{op}} \text{ The response is shown in 1.7}$$

1.1.2.6 Choice of Error Amplifier

We choose a single-stage amplifier because a two-stage amplifier will have extra capacitance, which introduces another pole.

In the paper [3], the author explains the benefits of folded cascode^{1.8}. This includes good input common mode range

Telescopic cascode^{1.9} often suffers from limited input/output range; in order to make output at its full swing, V_{in} and the bias voltages need to be carefully controlled, which is a big problem.

1.1.3 Decide for a topology which you will design in the following tasks. Explain your decision.

The topology we choose is 1.2 from Texas Instruments. It consists of only one stage amplifier. Compared to that is 1.10. This circuit has two stage amplifier, the high loop gain enables it to have better line and load regulation. It also introduced more pole and zeros, which makes it harder to reach stable, we need to do pole-zero cancellations. So the overall design will be more complex. Also we need to consider current consumption and circuit area. 1.2 is the simplest structure, any added circuit or component will require area and extra design complexity.

Table 1.5: Amplifier comparison

Topology	Telescope	Folded
Gain	Higher	Lower
Power consumption	Lower	Higher
Input/Output voltage swing	Narrower	Wider

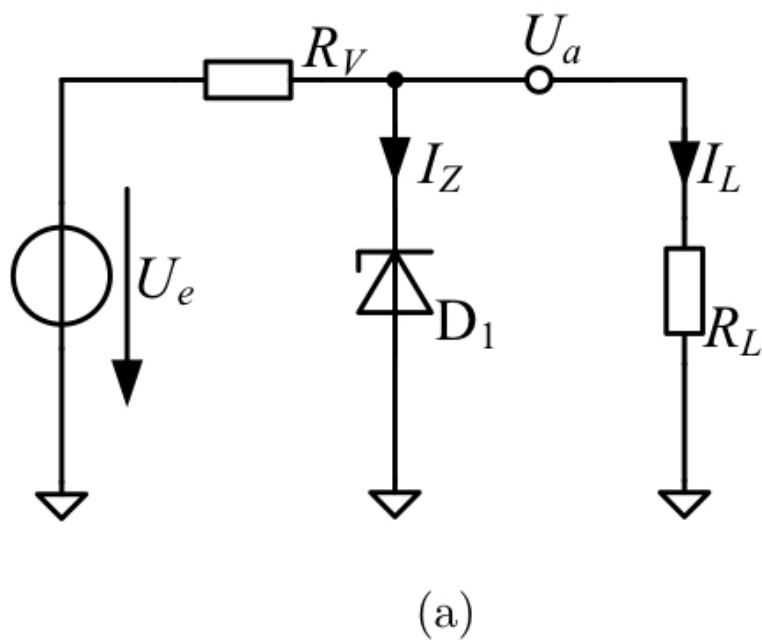


Figure 1.1: Example of an LDO without feedback

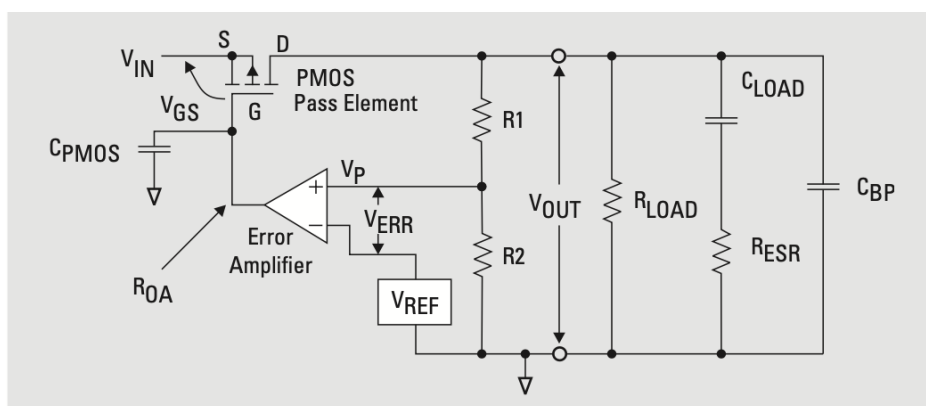


Figure 1.2: LDO schematic 1

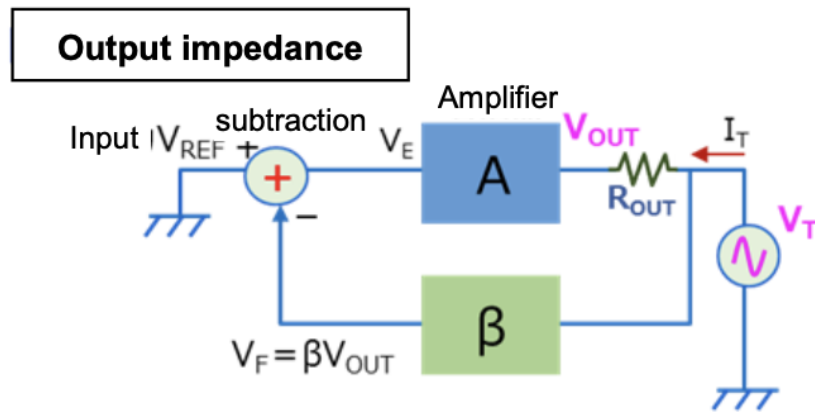


Figure 1.3: feedback

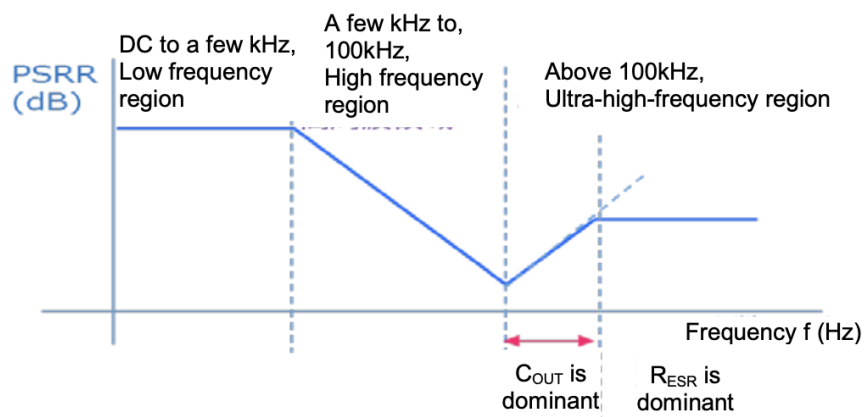


Figure 1.4: PSRR in different frequency

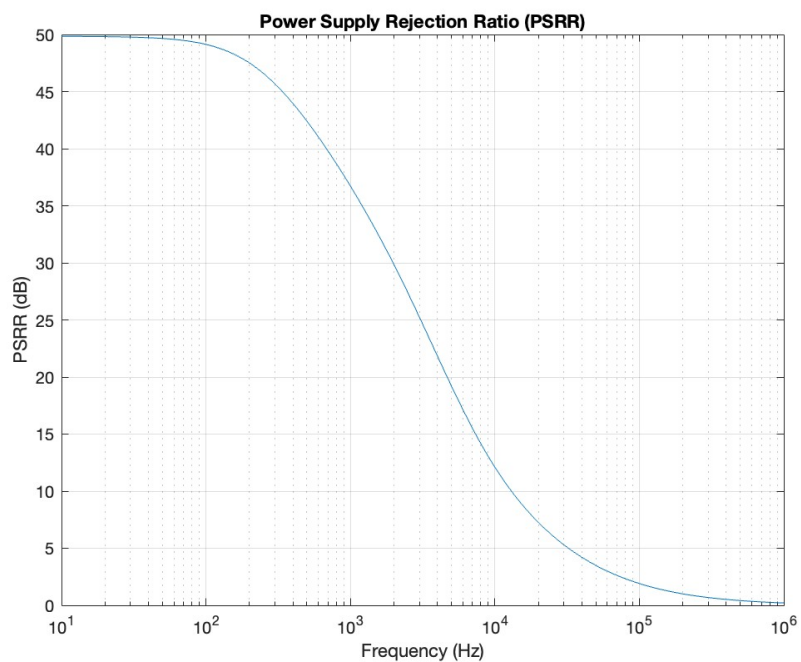


Figure 1.5: PSRR using my parameter

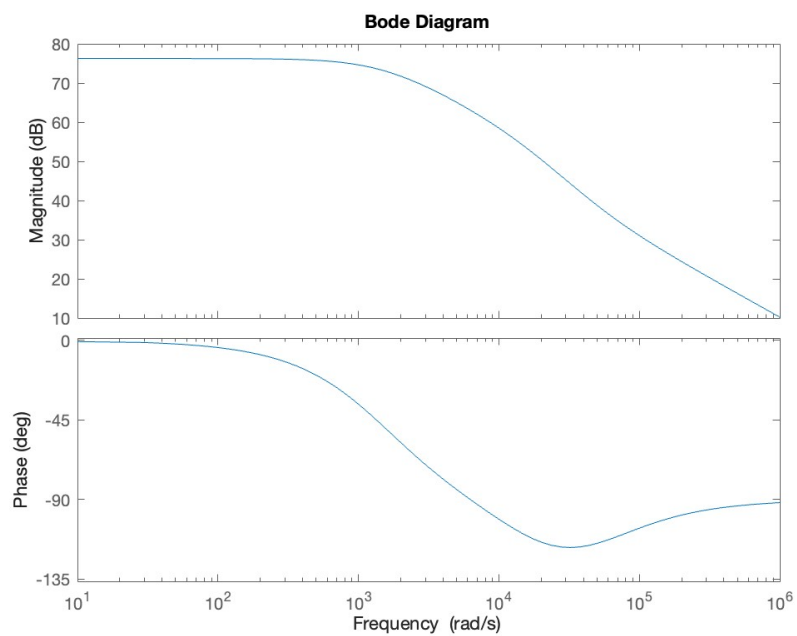


Figure 1.6: Bode plot for open-loop gain

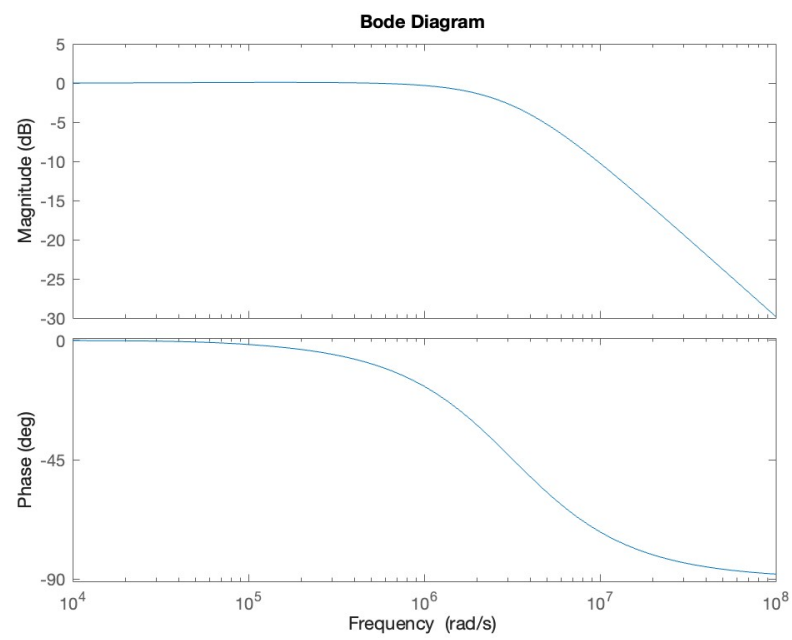


Figure 1.7: Bode plot for closed-loop gain



Figure 1.9: telescopic amplifier

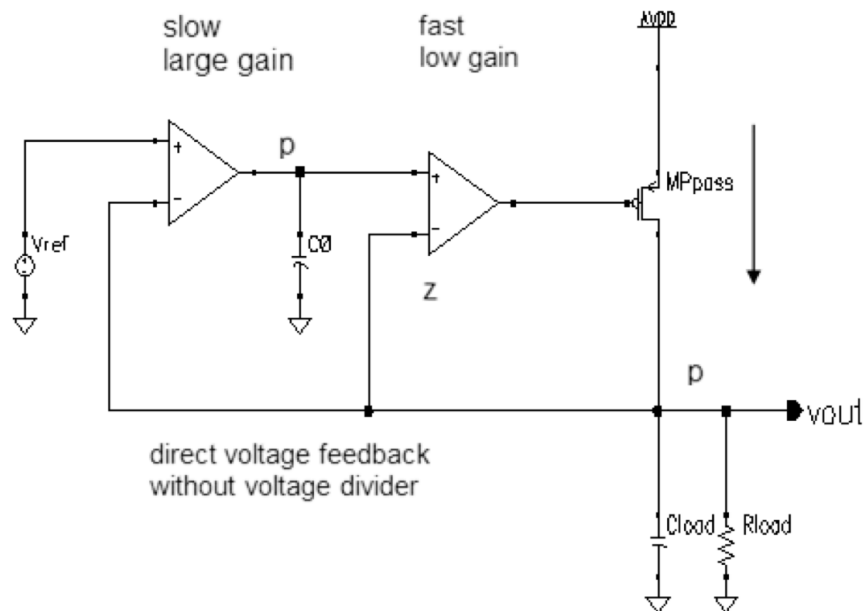


Figure 1.10: Structure 2 LDO

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1

1.1 Power regulator design questions

- What is the power supply rejection ratio (PSRR) of your circuit (with ideal OTA)?
- Using loop-gain measurements, extract the gain- and phase margin and make a statement on the stability of your circuit.
- What is the supply voltage dependence of V_{out} ? How will your circuit behave at high and low temperatures?
- Describe the settling behavior of the power supply when the load current changes and at start-up. Investigate different ramp-speeds!

Table 1.1: Specifications

Specification	Condition	Note
V_{DD}	1.5V–1.8V	Positive supply voltage
V_{SS}	0V	Negative supply voltage
V_{out}	$1.2V \pm 5\%$	Reference voltage
PSRR	$\geq 60\text{dB}$	Power supply rejection ratio
I_{load}	10 μ A–10mA	Current to be supplied
P	Minimize	$\leq 1\text{mW}$

1.1.1 What is the power supply rejection ratio (PSRR) of your circuit (with ideal OTA)?

PSRR is defined as:

$$\text{PSRR} = 20 \cdot \log \left(\frac{v_{inRPL}}{V_{outRPL}} \right)$$

Measure method: use xf analysis, apply 1V xf signal on VDD, then observe the effect on Vout.

The simulation of ideal OPAMP is figure1.5

In low frequency PSRR is improved by improving the loop gain (refer to [1]), in our simulation result the PSRR in low frequency is very good 100dB. As frequency goes higher, the gain of OPAMP drop, PSRR will go up. In higher frequency, we have to consider our load capacitance, so PSRR goes up again.

1.1.2 Using loop-gain measurements, extract the gain and phase margin and make a statement on the stability of your circuit.

Measurement method: Use stb analysis.

The loop gain at low frequency is 126 dB, and the pole and zero locations are explained in [2]. Our structure is shown in Figure 1.1. Below are our parameters.

In our waveform, we first observe the dominant pole is at approximately 10 Hz. We can move it to the right by making C_{load} smaller; however, C_{load} is important to stabilize our circuit when I_{out} changes. Thus, C_{out} cannot be too small. The first zero is at approximately 5 kHz. The phase margin is the distance to -180 degrees at unity gain. Our circuit has a 90-degree phase margin, indicating stability.

Table 1.2: Parameters

Parameter	Value
PMOS(W/L)	100 μ m/150nm
V_{ref}	1.1V
V_{out}	1.2V
R_1, R_2	1k Ω , 11k Ω
C_{load}	10 μ F
R_{ESR}	2 Ω
I_{out}	10 μ A to 10mA

1.1.3 What is the supply voltage dependence of V_{out} ? How will your circuit behave at high and low temperatures?

Refer to Figure 1.6. As the supply voltage rises from 1.5 V to 1.8 V, we can observe that V_{out} becomes stable at 1.68 V. This is the point where regulation starts.

When the temperature is low, V_{out} is constant at 0.6 V. When the temperature rises, V_{out} will decline. This is because the threshold voltage (V_{th}) of the PMOS decreases, causing it to enter the linear region. As a result, the drain current (I_D) drops, leading to a decrease in the output voltage.1.3

1.1.4 Describe the settling behavior of the power supply when the load current changes and at start-up. Investigate different ramp speeds!

To observe the settling behavior, we apply a unity step function to I_{out} , changing it from 10 μA to 10 mA.

When the current is too low, the gate voltage of our PMOS needs to be higher to reduce the current. However, the maximum voltage is limited by v_{soft} , which is defined as the gap between V_{supply} and V_{out} of the error amplifier. We make v_{soft} smaller to allow V_{out} to reach a higher value.

It is better for the PMOS to be smaller in size so that the capacitance and area are reduced. By making the PMOS smaller, the V_{gs} will be higher, and the V_{out} of the error amplifier can be lower. v_{soft} can be larger, which is closer to a real OPAMP.

The requirement is $1.2\text{ V} \pm 5\%$, which is 1.14 V to 1.26 V. As seen in Figure 1.2, our output is within this range.

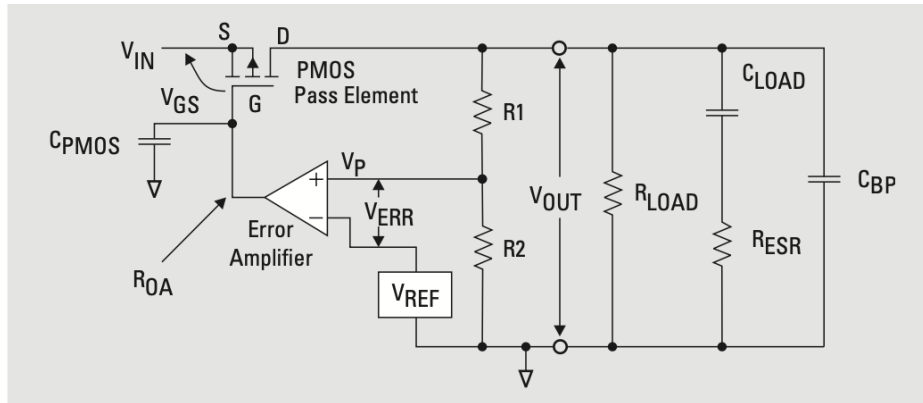


Figure 1.1: LDO structure

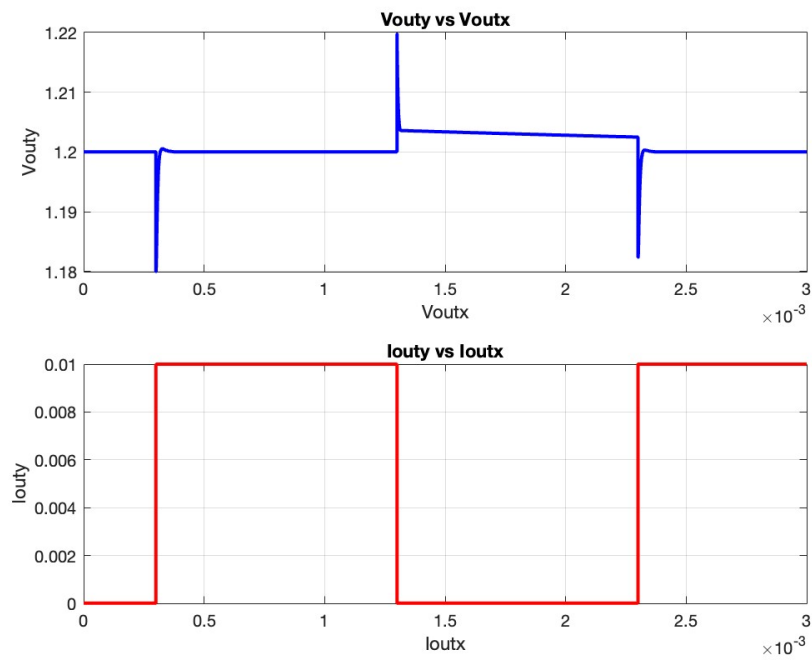


Figure 1.2: settling time for pulse input in load current

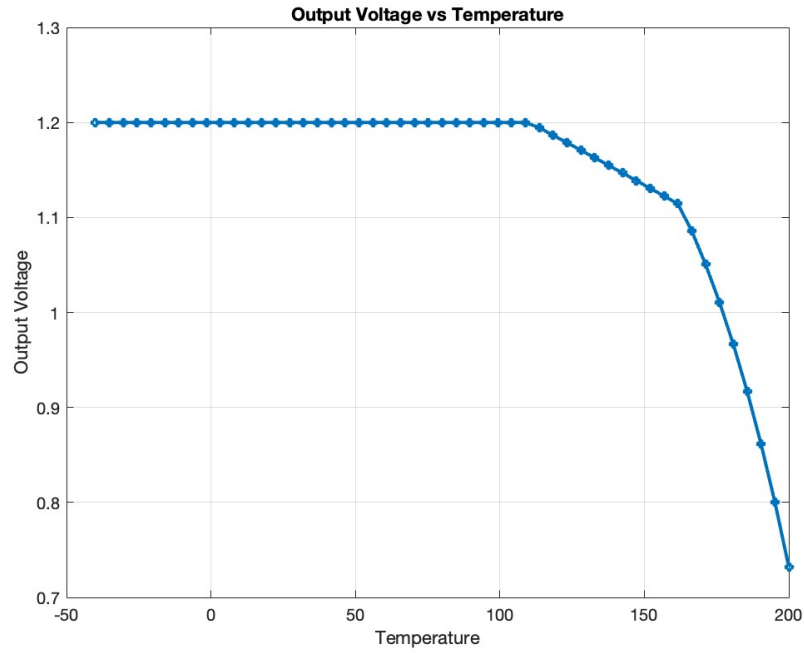


Figure 1.3: Vout under different temperature

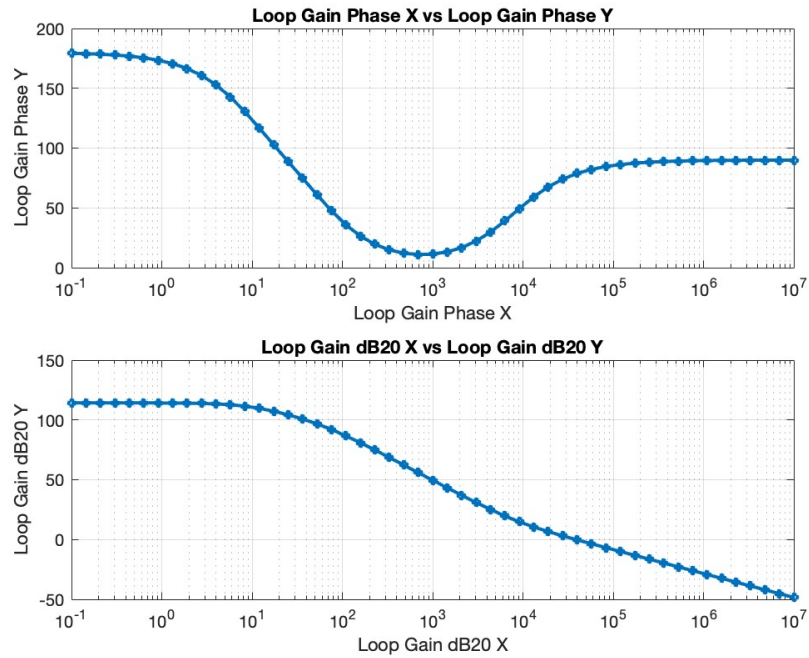


Figure 1.4: Loop gain in dB 20 and phase margin

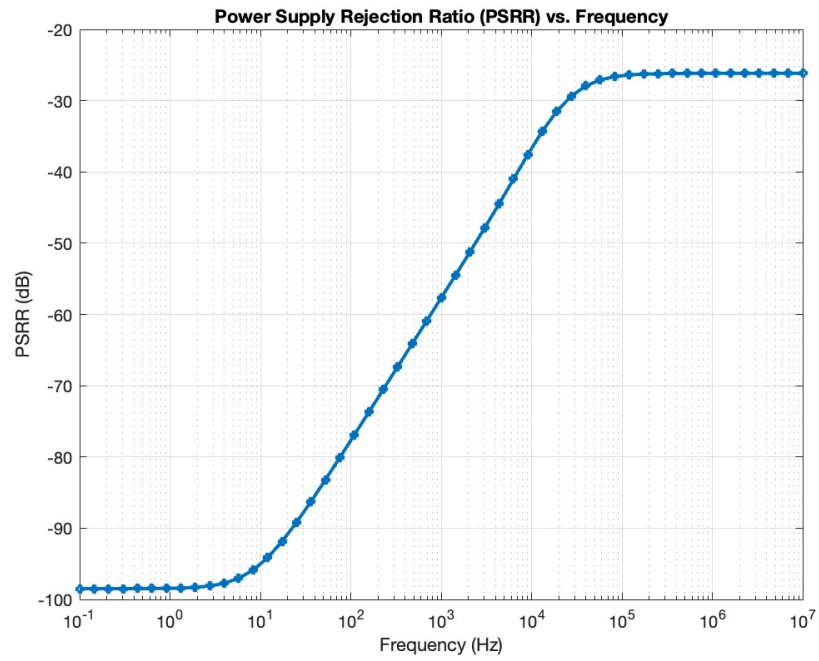


Figure 1.5: Ideal PSRR

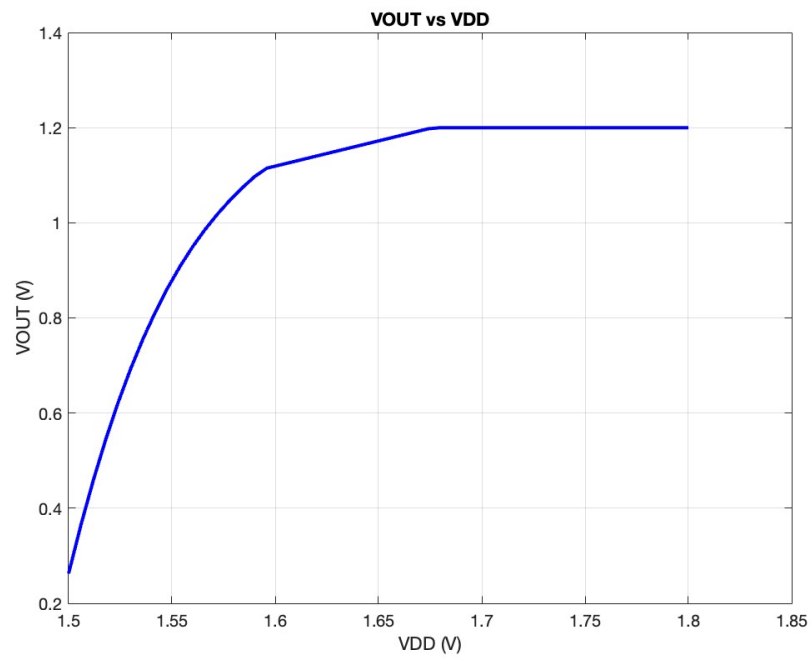


Figure 1.6: VDD vs Vout

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Lab Report

CMOS Voltage Regulation Circuit design

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July 19, 2024

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1

1.1 Power Regulator Design Questions

- Perform the same characterizations as before. How do they compare to the ideal case (task 3)?
- Now assume a $5 \text{ mV} \cdot \mu\text{m}$ input (or V_{TH}) mismatch (add an equivalent voltage source to simulate the effect) just between the input transistors of your first amplifier and describe how this affects your circuit's worst case.
- Assume a dynamic load jump from min to max supply current and vice versa: How fast is your circuit reacting (settling to steady state)?
- What happens if your circuit is in the low current state and the output sees a small negative transient current?
- What happens when the supply voltage is gradually ramped from 0V to 1.8V?
- Run a simulation including the process corners (FS, FF, SF, SS). Does your final circuit still meet the specification in every PVT corner?
- Run a Monte-Carlo Simulation (100 samples, DC simulation only) including mismatch and process variations on all devices of your final circuit. How many samples meet the specifications?

1.1.1 Amplifier Design

The amplifier we are using (see Figure 1.2) is a folded cascode amplifier. Folded cascode amplifiers, as referenced in [1], are known for providing high gain and a good input range. We have chosen NMOS as our input. Normally, NMOS transistors have higher transconductance (g_m), which is beneficial for achieving high gain. However, in our technology (gpdk045), both NMOS and PMOS transistors have the same g_m , so the choice of transistor type does not affect gain directly.

The choice of input transistor type also depends on the input DC voltage. In our design, the input DC voltage is 1.1 V. If we were to use a PMOS input pair instead of NMOS, the headroom for the input voltage ($1.8 \text{ V} - V_{ds_current_mirror}$) would be smaller.

We have opted for a single-stage amplifier due to its ease of stability. Although a two-stage amplifier with Miller compensation could potentially improve gain slightly, the additional pole introduced by the second stage often results in a worse phase margin. The gain of our amplifier is approximately $\frac{1}{2}g_m^2 r_o^2$.

The folded cascode stage consumes more power, and there are various trade-offs in amplifier sizing. For instance, starting with larger values of width (W) and length (L) for MOSFETs can mitigate mismatch effects. Moreover, larger MOSFET sizes typically reduce flicker noise. The g_m/I_D method is useful for sizing the amplifier effectively.

Table 1.1: Specifications

Specification	Ideal case	Real Implementation
VDD	1.8V	1.8V
VSS	0V	0V
Vout	1.2V	1.2V
PSRR	-100dB	-63.11dB
Gain/ Phase Margin	130dB/ 88 deg	80dB/ 67.8 deg
Settling voltage	$1.2\text{V} \pm 5\%$	1.18 - 1.22 V (within 5%)

1.1.2 Real implementation compared to ideal OPAMP implementation

Refer to figure 1.4 and 1.5, we can see a PSRR higher than 62 dB, and phase margin 67 degree. Compared to ideal implementation, where PSRR is 100 dB in low frequency, real implementation has lower amplifier gain, so PSRR is lower.

1.1.3 Now assume a $5 \text{ mV} \cdot \mu\text{m}$ input (or V_{TH}) mismatch (add an equivalent voltage source to simulate the effect)

Refer to Figure 1.27. Assume the V_{TH} of two input MOSFETs don't match. We can put a small voltage source on one input node to simulate. The voltage we add should

Table 1.2: Parameters

Parameter	Value
PMOS(W/L)	100 μ m/150nm
V_{ref}	1.1V
V_{out}	1.2V
R_1, R_2	1k Ω , 11k Ω
C_{load}	10 μ F
R_{ESR}	2 Ω
I_{out}	10 μ A to 10mA

be proportional to $\frac{5\text{ mV}}{\sqrt{L \cdot W}} = \frac{5\text{ mV}}{\sqrt{1\mu\text{m} \cdot 8\mu\text{m}}} = 1.76\text{ mV}$. The output will also go up because the resistive voltage divider will multiply the small offset voltage by $\frac{12}{11}$.

In our figure, we sweep the small voltage source from 0 to 2 mV. We can see that the output voltage is still within $\pm 5\%$.

1.1.4 Assume a dynamic load jump from min to max supply current and vice versa: How fast is your circuit reacting (settling to steady state)?

When the load current jumps from low to high, we will see a negative pulse in V_{out} . According to paper [2], this is because of the equivalent series inductance in our C_{load} . When the current increases suddenly, there will be a negative drop across the inductance. Additionally, there is series resistance in our C_{load} ; together with R_2 , they will cause a voltage drop when the current suddenly increases. As the PMOS starts to provide more current, our output voltage will return to the regulated value of 1.2V. To minimize the problem, we want a smaller R_2 , but R_2 is also related to loop stability. To mitigate the effects of inductance and internal resistance of C_{load} , we can use different types of capacitors.

From Figure 1.7 and Figure 1.8, we can see that the settling time for rising edge is approximately 70ns, while the settling time for falling edge is approximately 100ns. Both are very fast. [3] provides the formula for settling time. In order to make it faster, we have to improve our closed loop bandwidth and minimise the output capacitance of EA.

1.1.5 What happens if your circuit is in the low current state and the output sees a small negative transient current?

1.6 shows that if there's a small negative current, the output voltage will have a spike because more current flow in the resistors, shortly after the current will return to 1.2V if negative current disappear.

1.1.6 What happens when the supply voltage is gradually ramped from 0V to 1.8V?

From 1.9 we can see that when supply voltage is higher than approximately 1.4V, the LDO start to regulate. This is because when supply voltage is too low our PMOS cannot provide enough load current.

1.1.7 Run a simulation including the process corners (FS, FF, SF, SS). Does your final circuit still meet the specification in every PVT corner?

We first measure gain and phase margin under 4 corners. 1.10 1.11 1.12 1.13, the phase margin are all > 60 dB, stable.

Then we measure PSRR, the requirement is > 60 dB. From 1.14 1.15 1.16 1.17 we can see that all of them are above 60dB.

Finally we measure the settling time under different corners. The results are 1.18 1.19 1.20 1.21 1.22 1.23 1.24 1.25

1.1.8 Run a Monte-Carlo Simulation (100 samples, DC simulation only) including mismatch and process variations on all devices of your final circuit. How many samples meet the specifications?

The Monte Carlo result is shown in Fig. 1.26. As can be seen, all the values fall within the range $1.2\text{ V} \pm 5\%$.

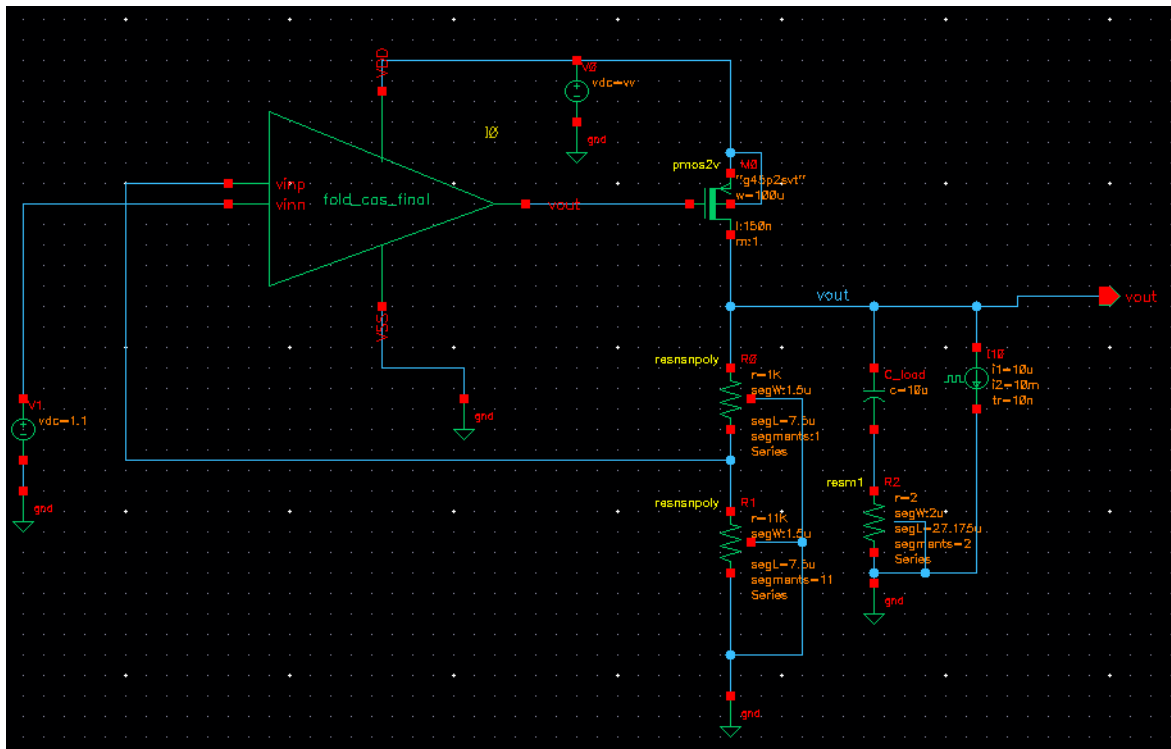


Figure 1.1: LDO schematic

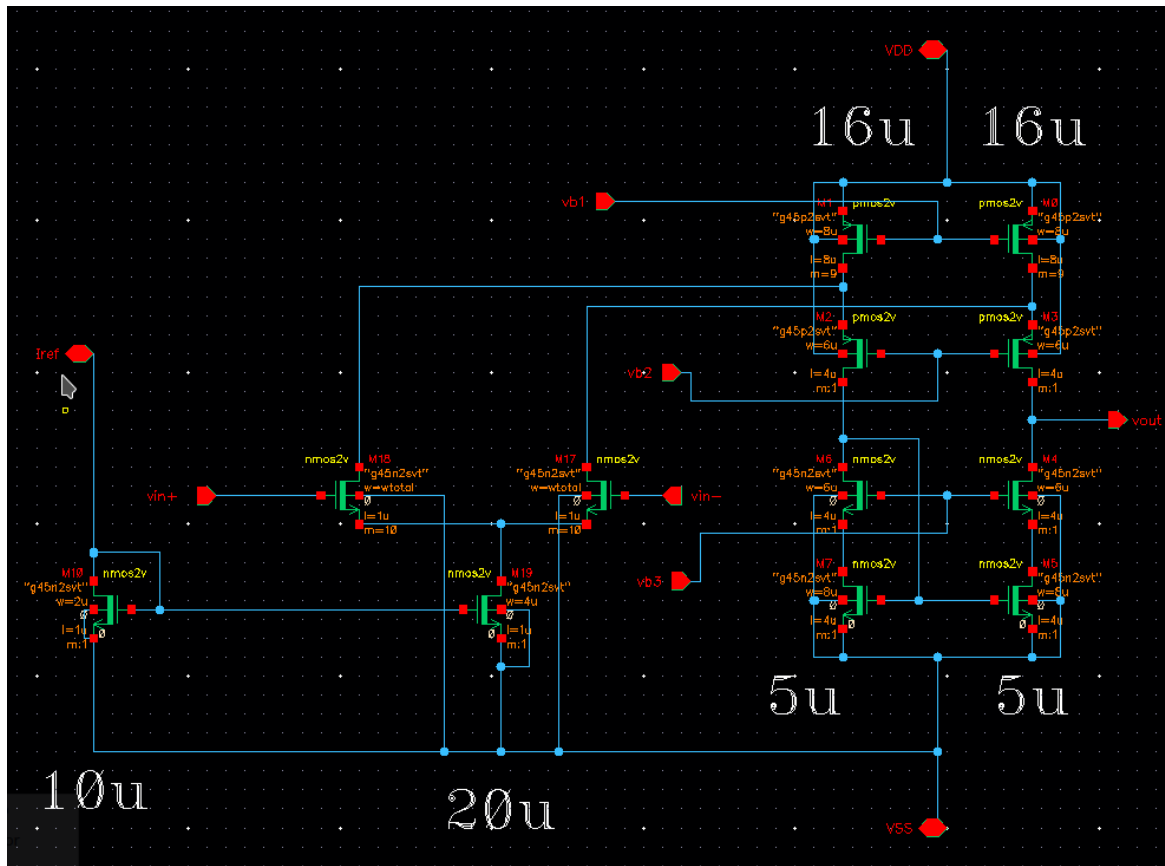


Figure 1.2: Folded cascode amplifier

1.1 – POWER REGULATOR DESIGN QUESTIONS

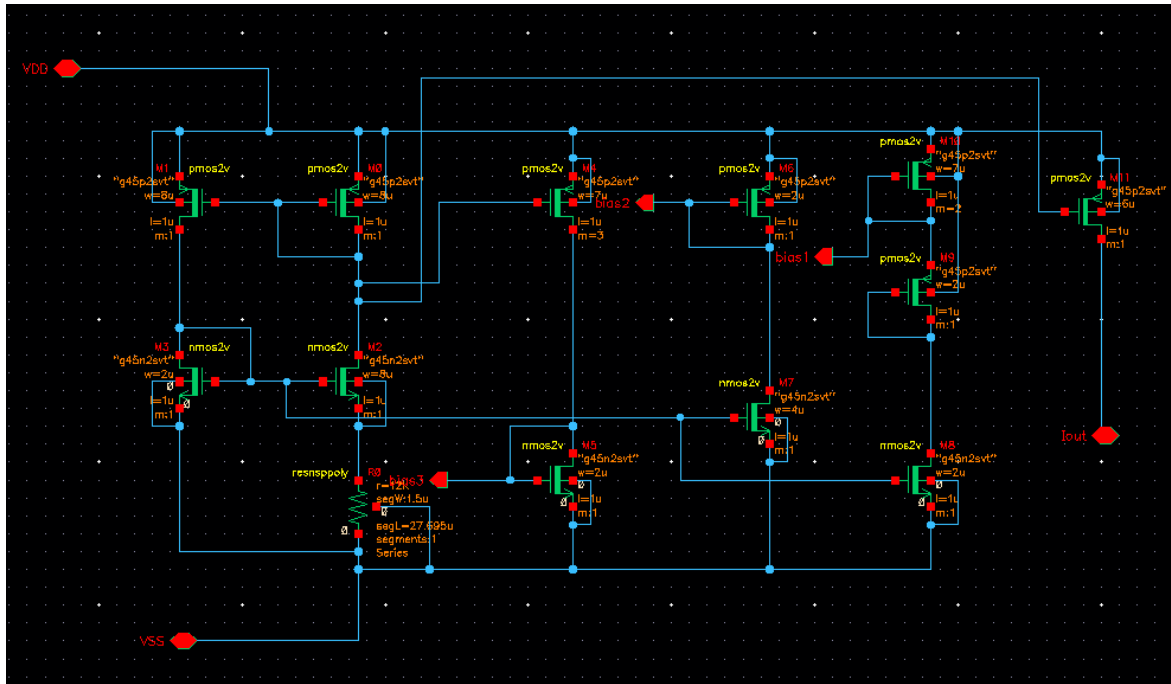


Figure 1.3: Biasing circuit

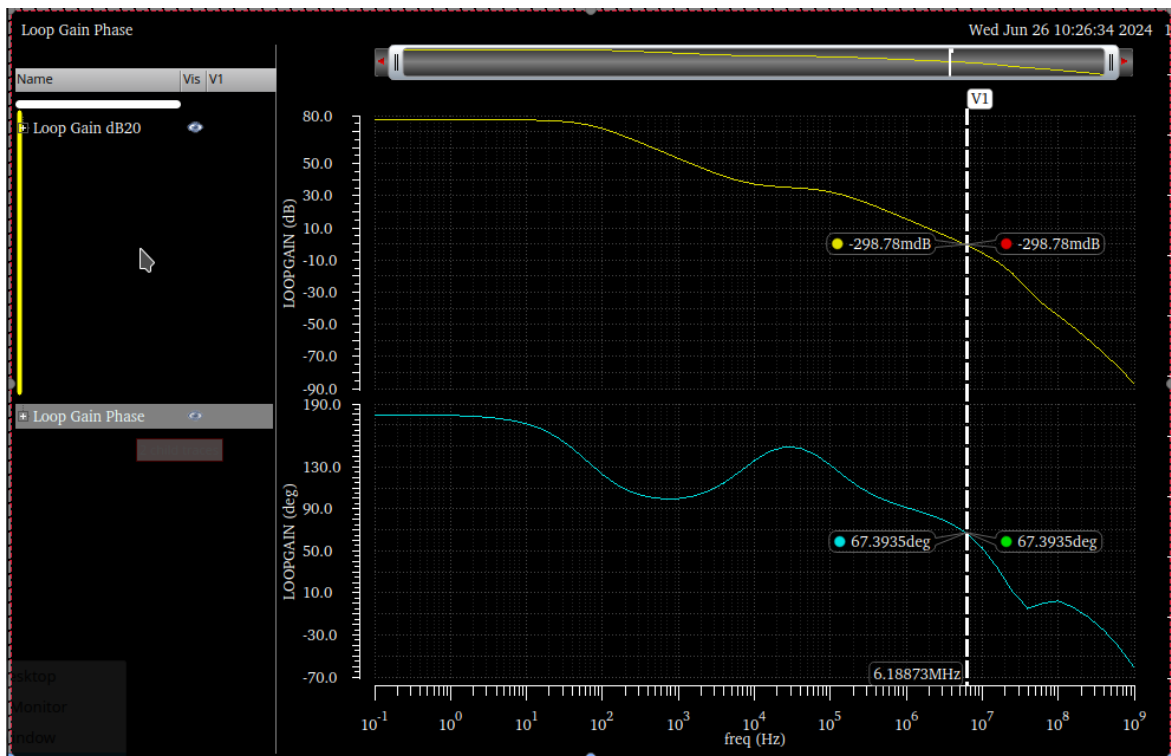


Figure 1.4: Gain, Phase Margin in TT

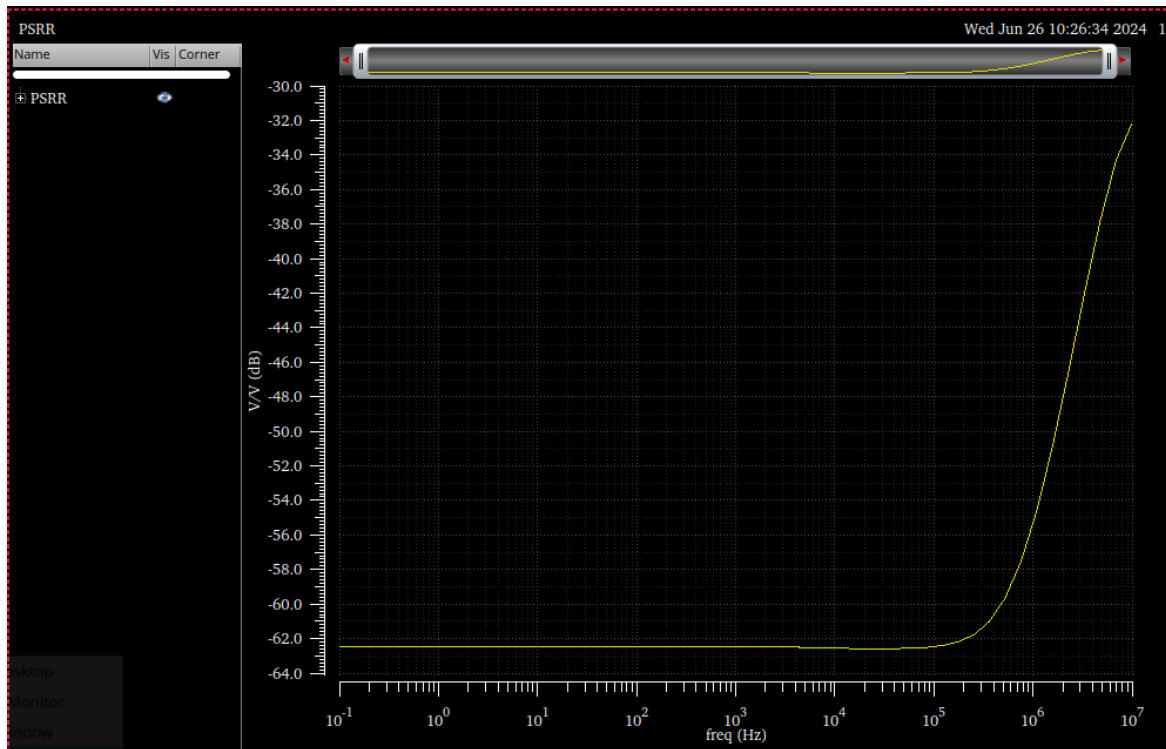


Figure 1.5: PSRR in TT

1.1 – POWER REGULATOR DESIGN QUESTIONS

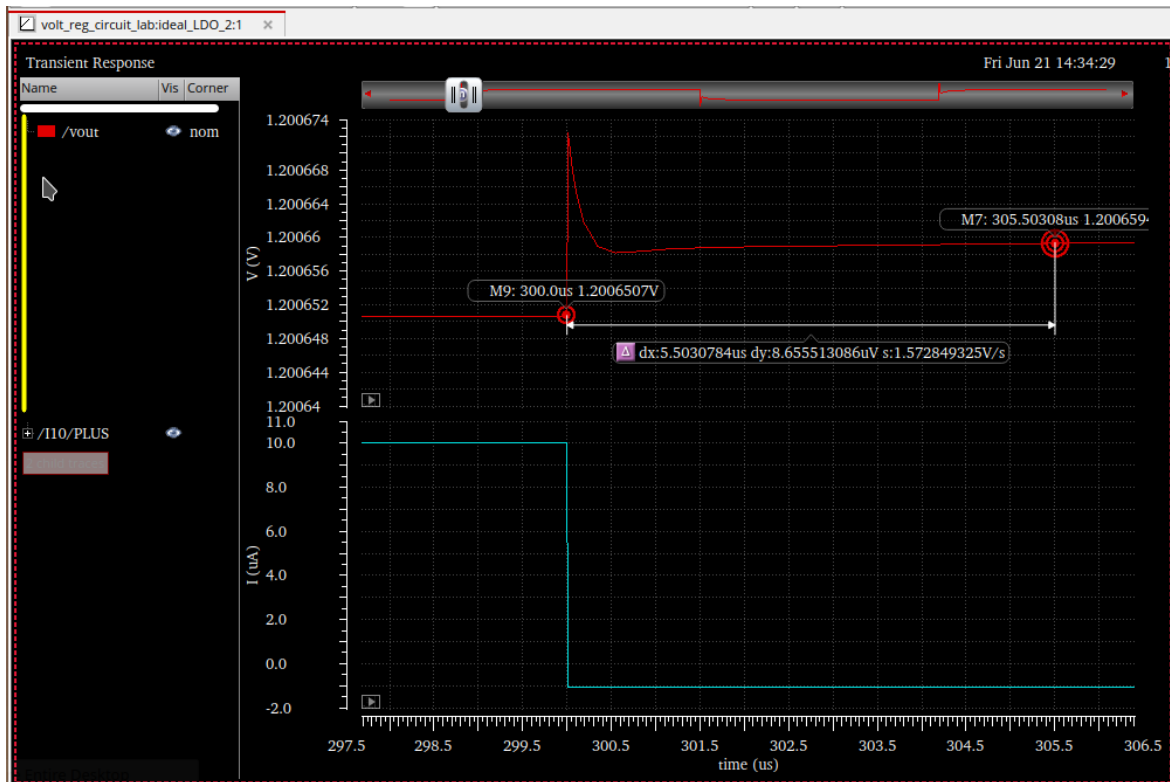


Figure 1.6: negative transient current

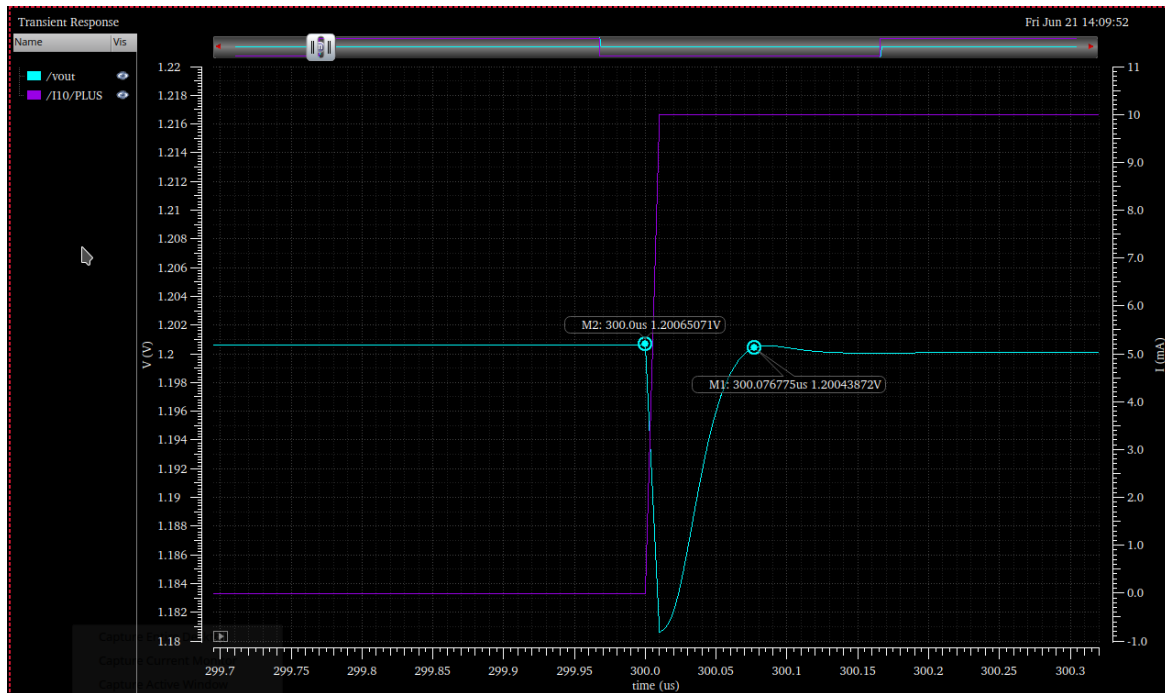


Figure 1.7: Output current jump low -> high

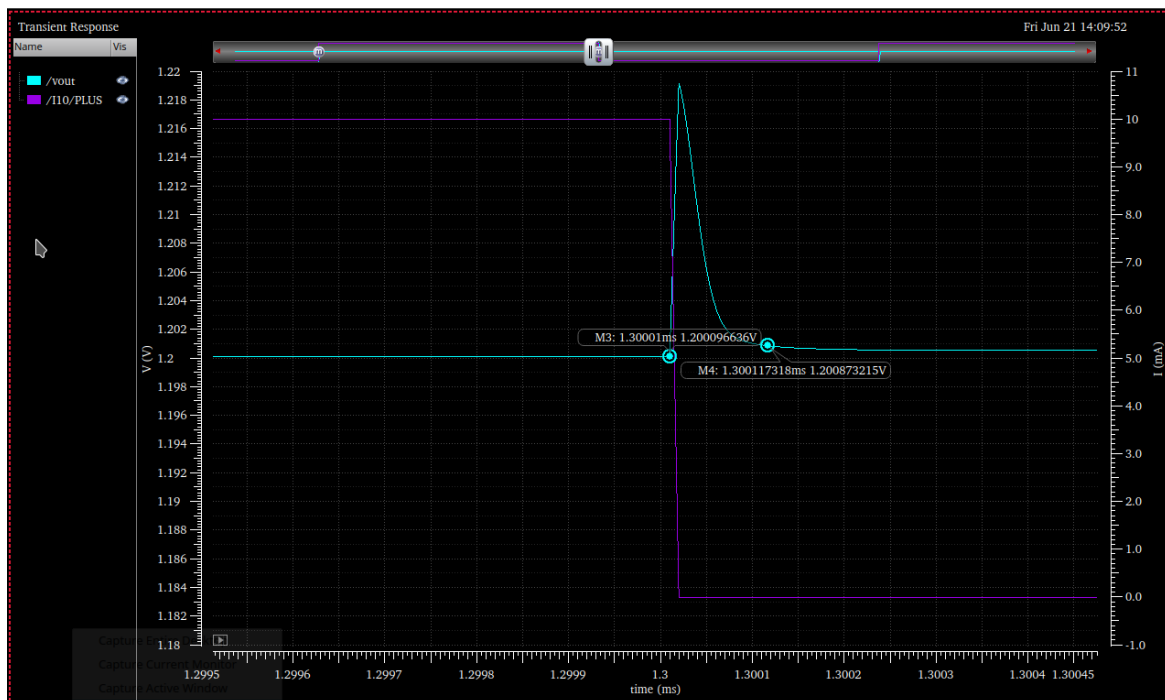


Figure 1.8: Output current jump high -> low

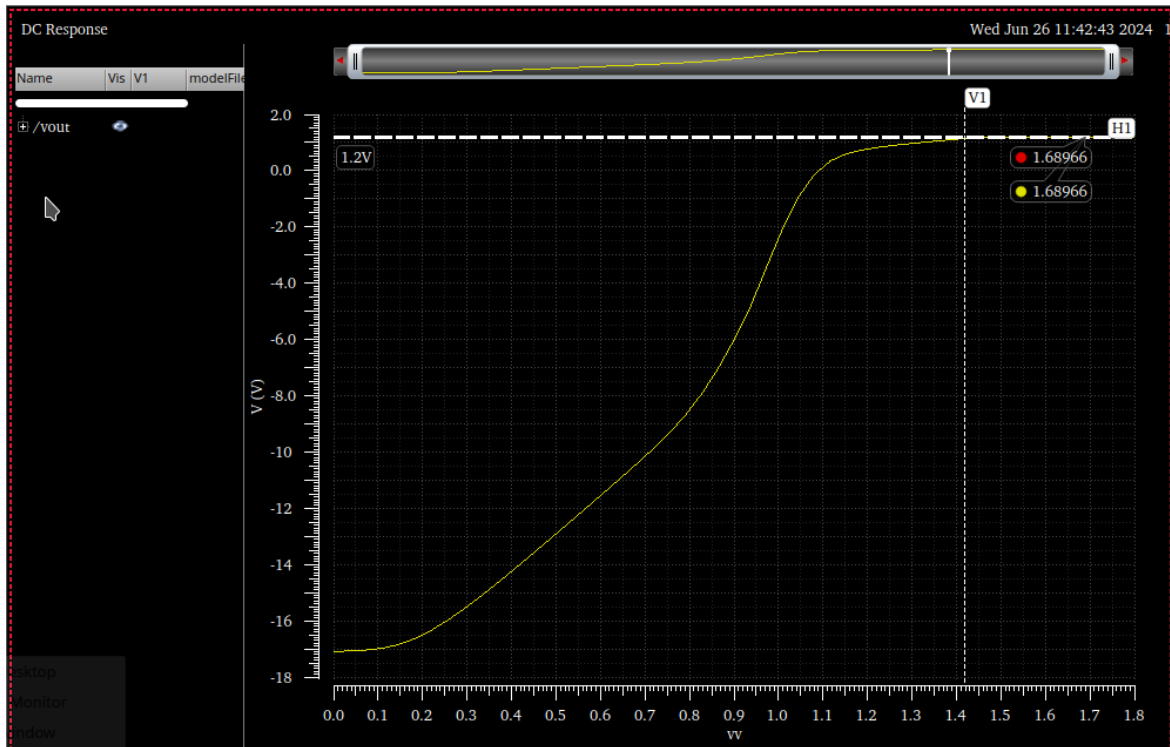


Figure 1.9: VDD ramp from 0 to 1.8V

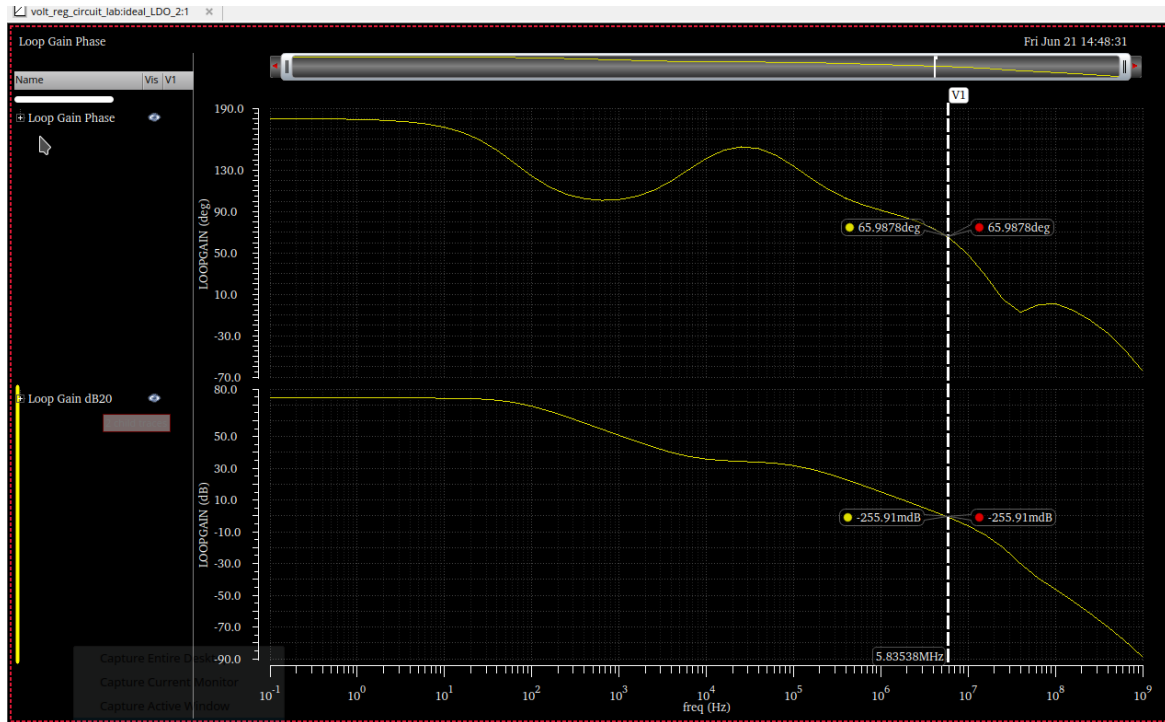


Figure 1.10: Gain, Phase Margin FS

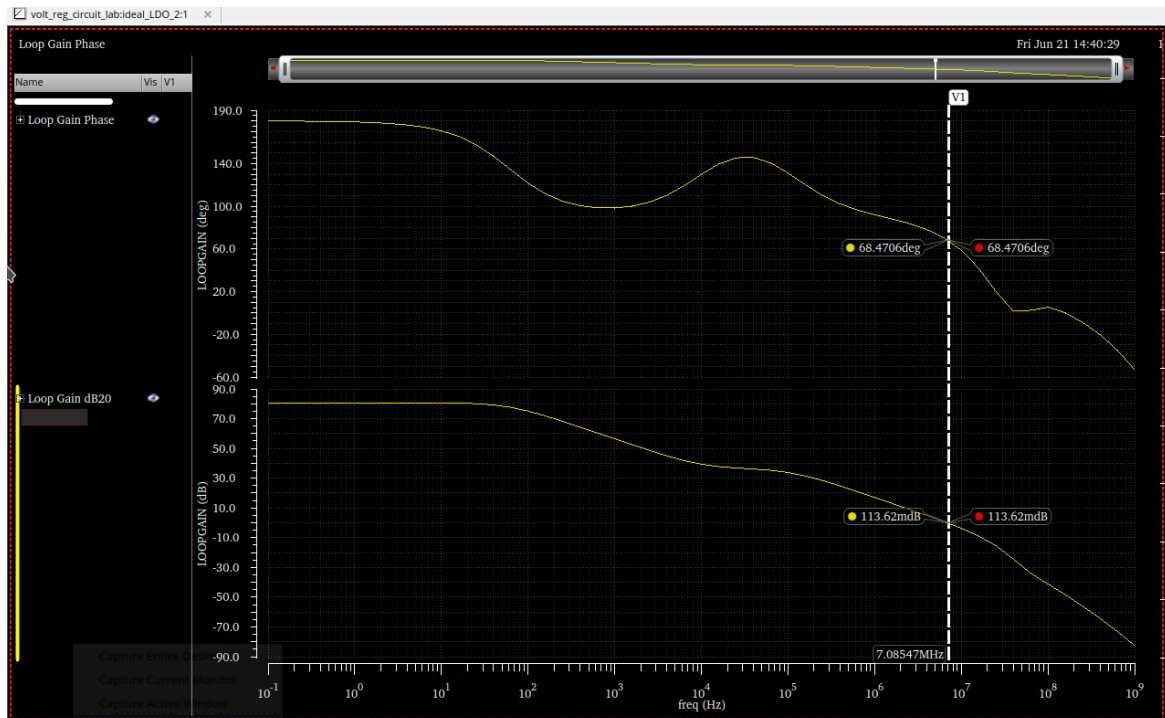


Figure 1.11: Gain, Phase Margin FF

1.1 – POWER REGULATOR DESIGN QUESTIONS

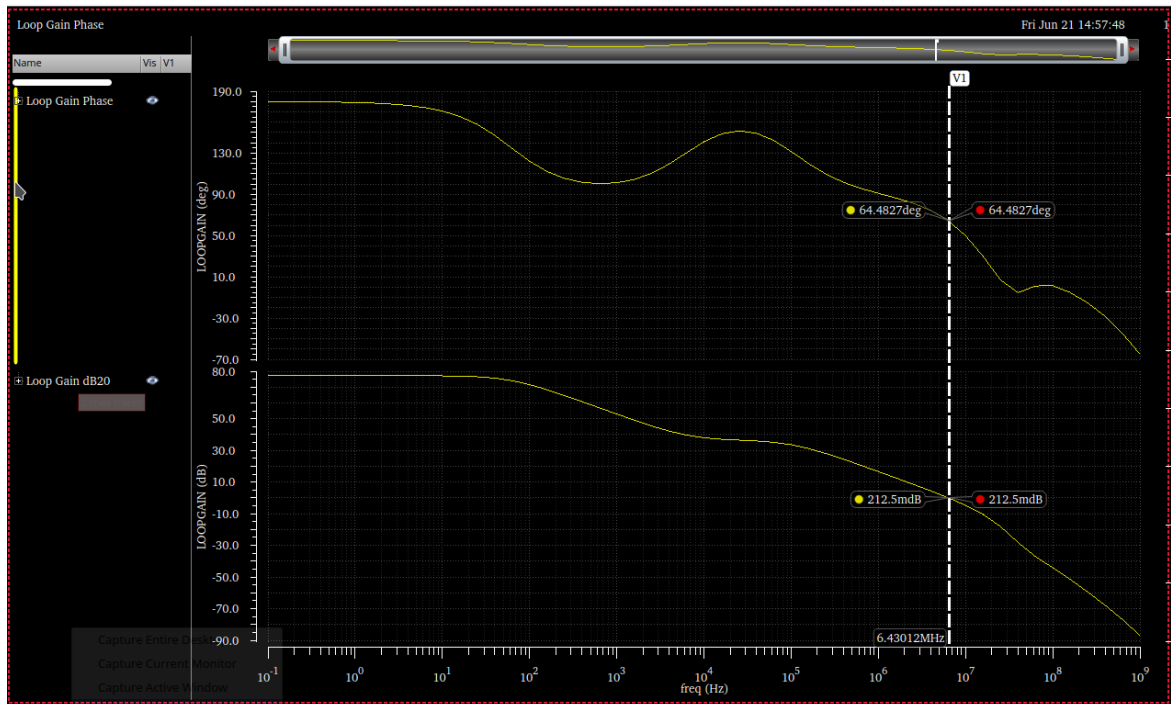


Figure 1.12: Gain, Phase Margin SF

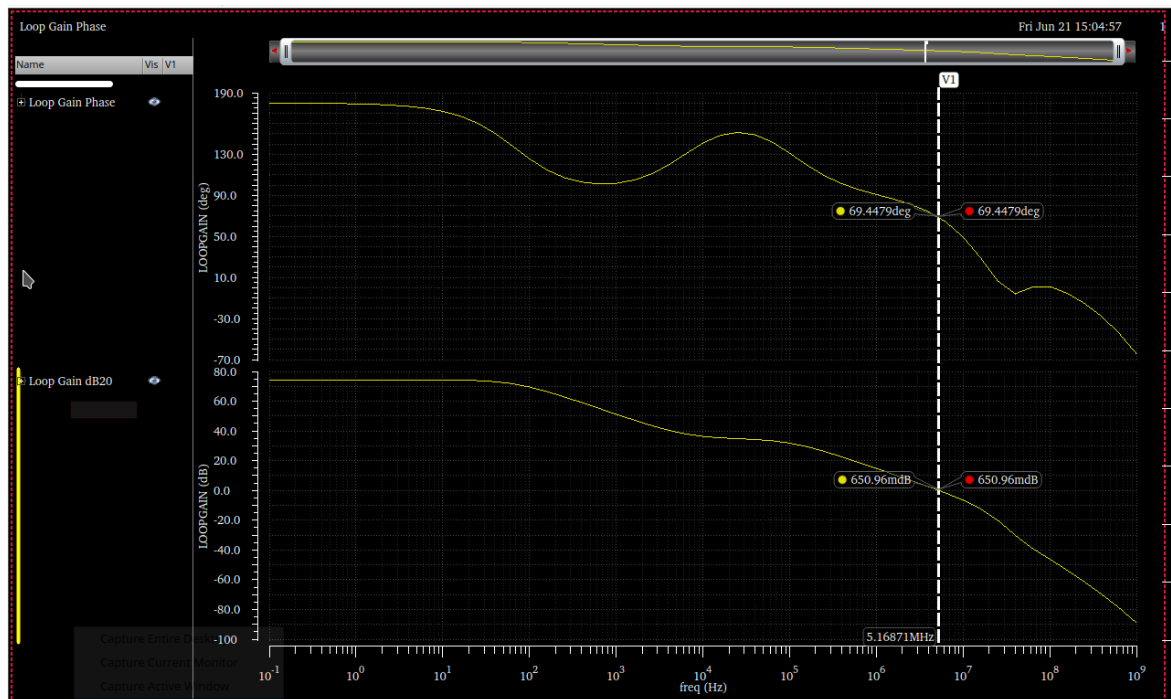


Figure 1.13: Gain, Phase Margin SS

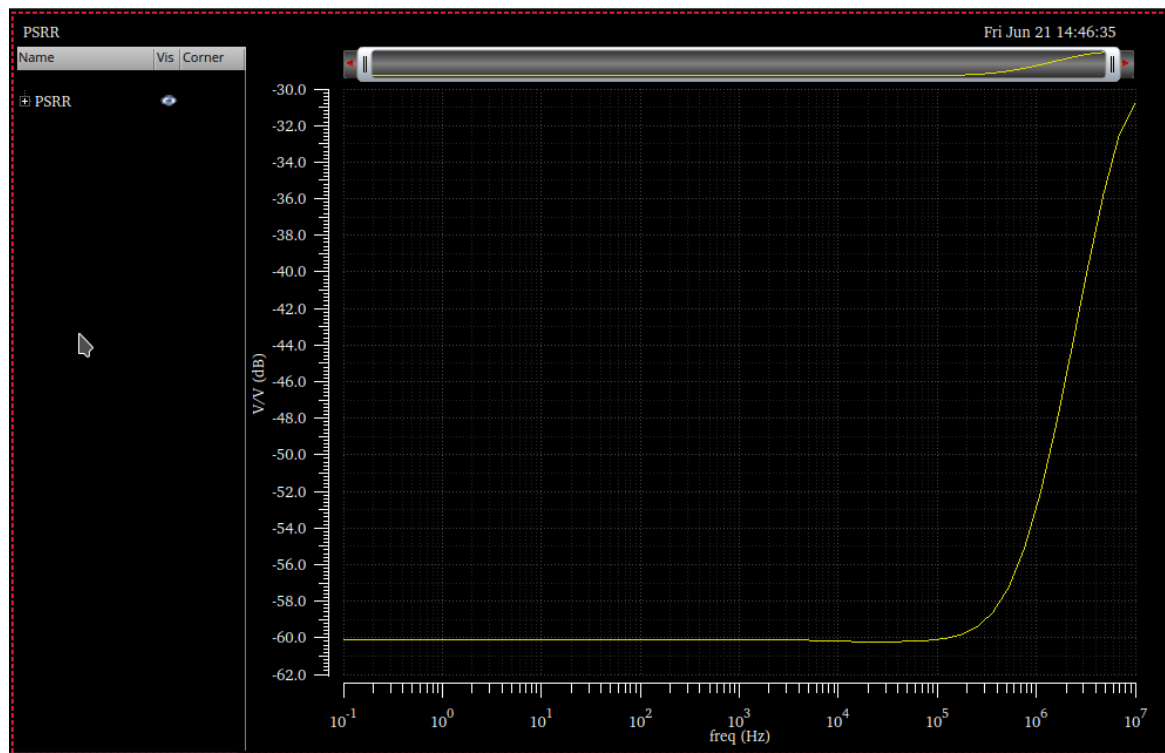


Figure 1.14: PSRR FS

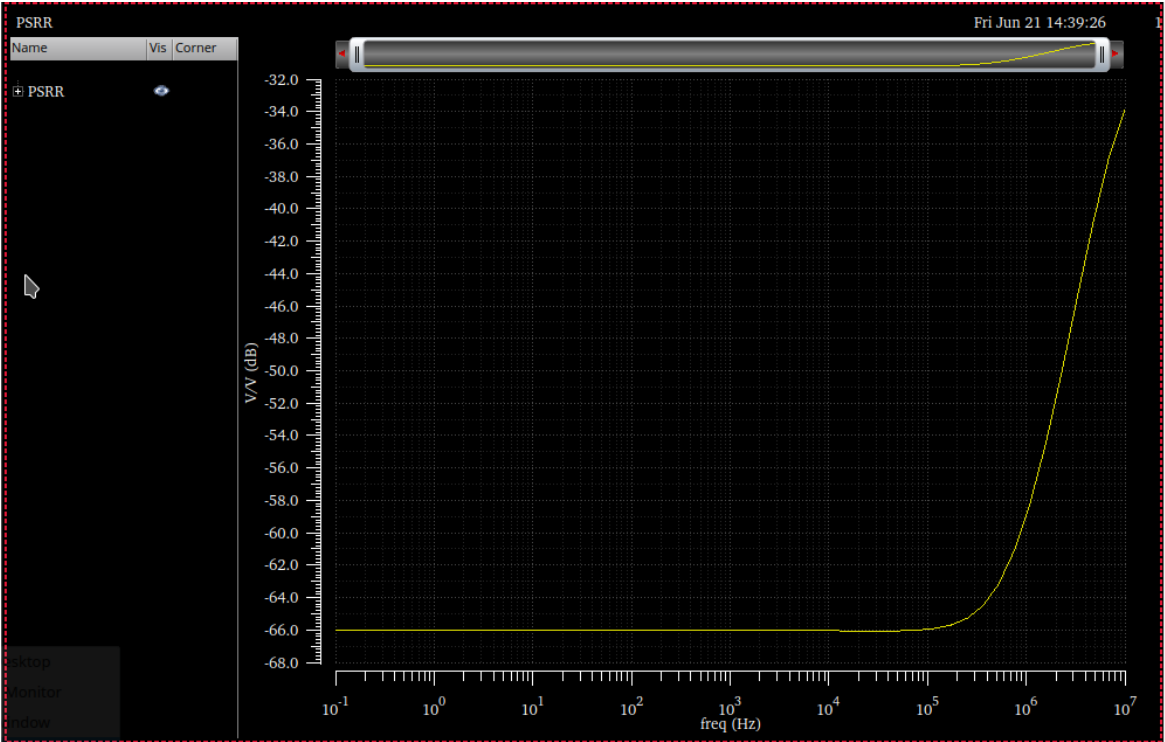


Figure 1.15: PSRR FF

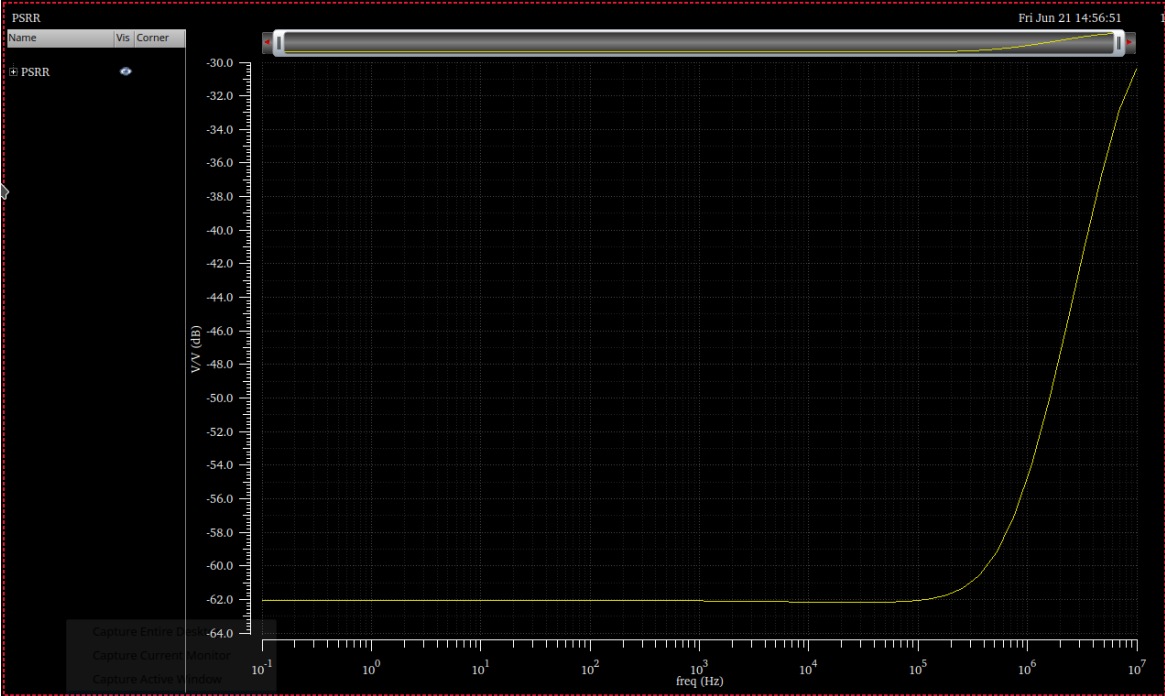


Figure 1.16: PSRR SF

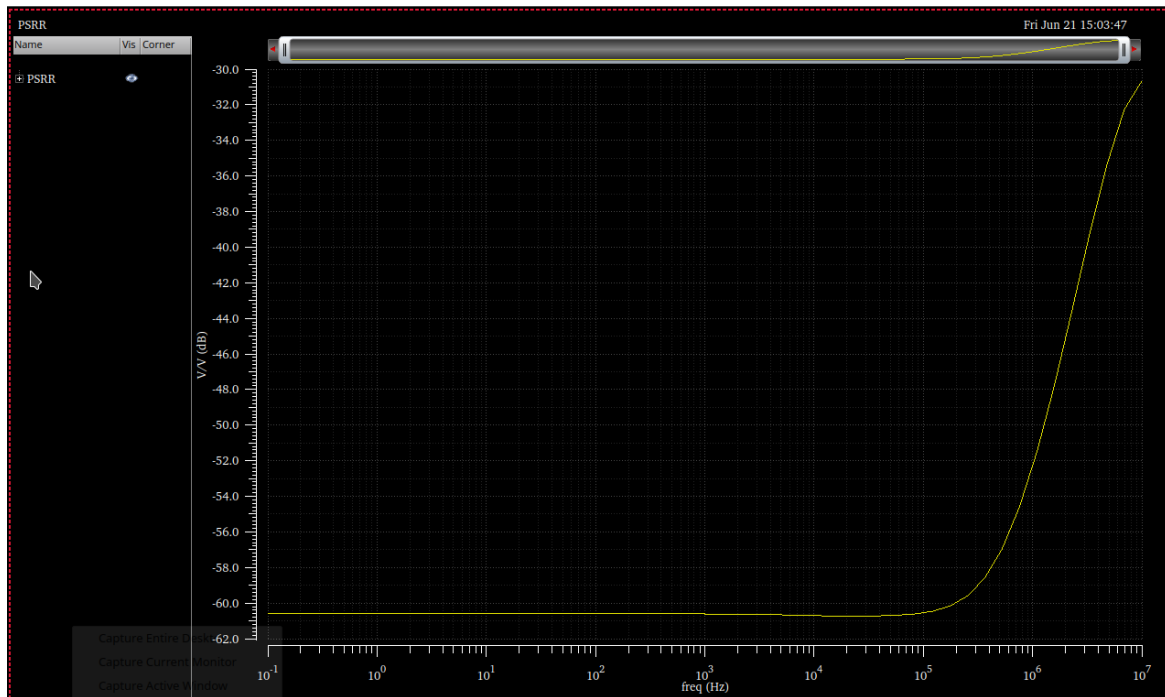


Figure 1.17: PSRR SS

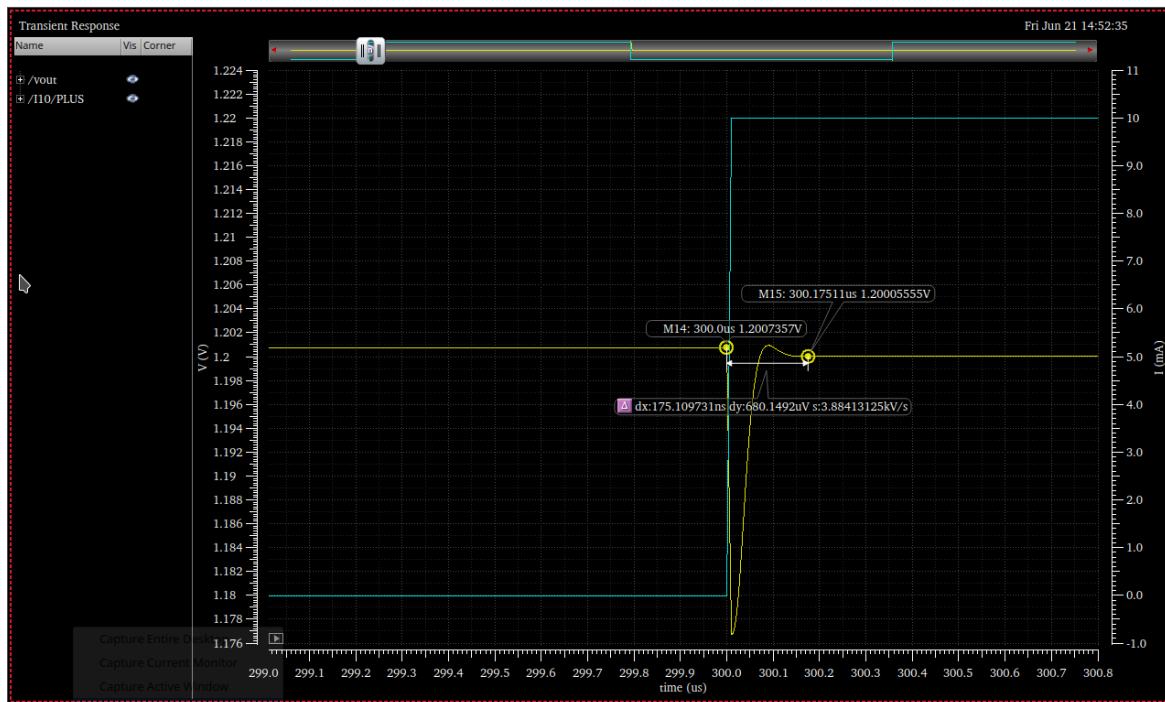


Figure 1.18: settling time rising FS

1.1 – POWER REGULATOR DESIGN QUESTIONS

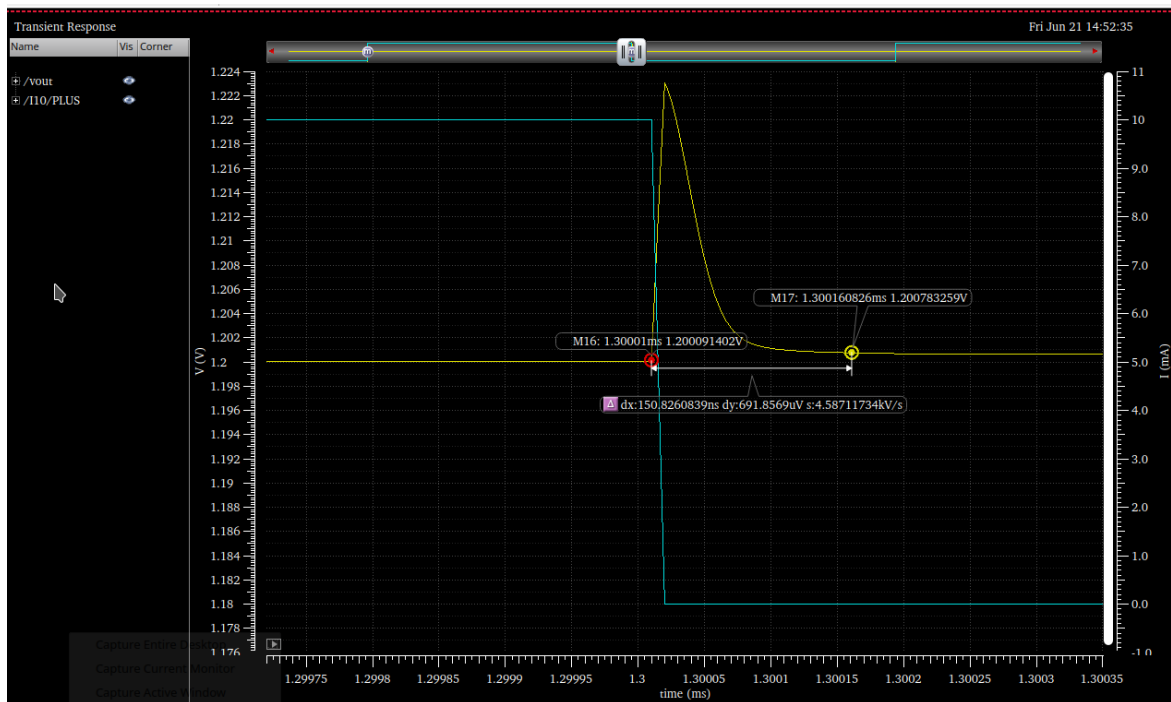


Figure 1.19: settling time falling FS

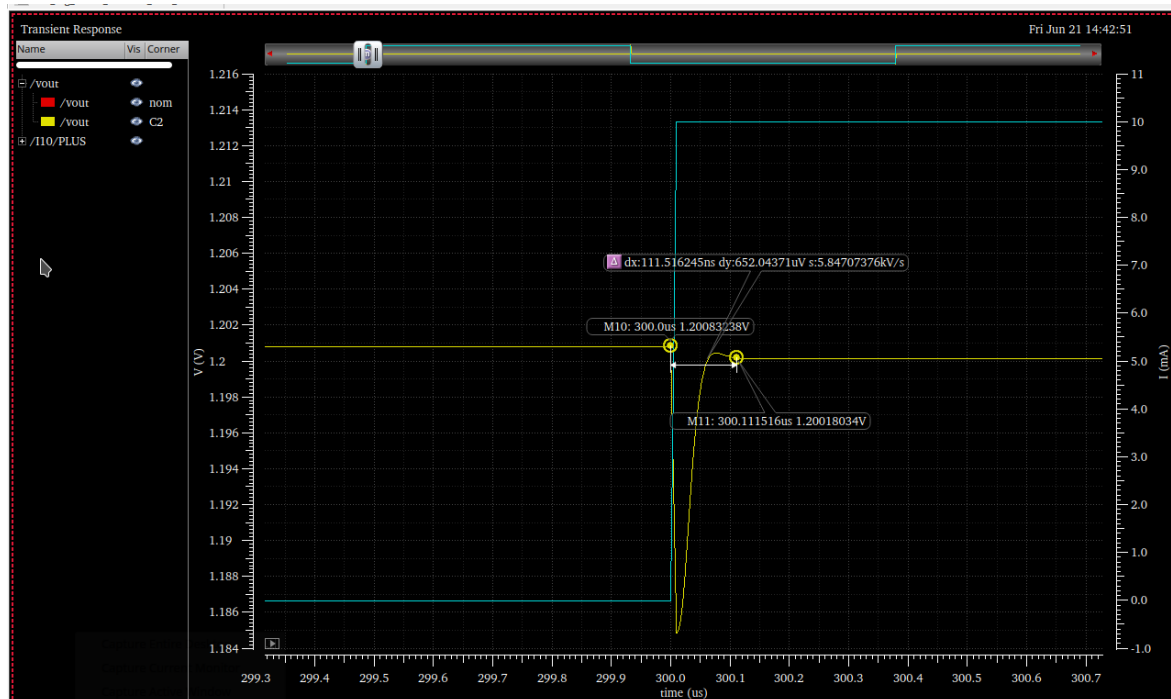


Figure 1.20: settling time rising FF

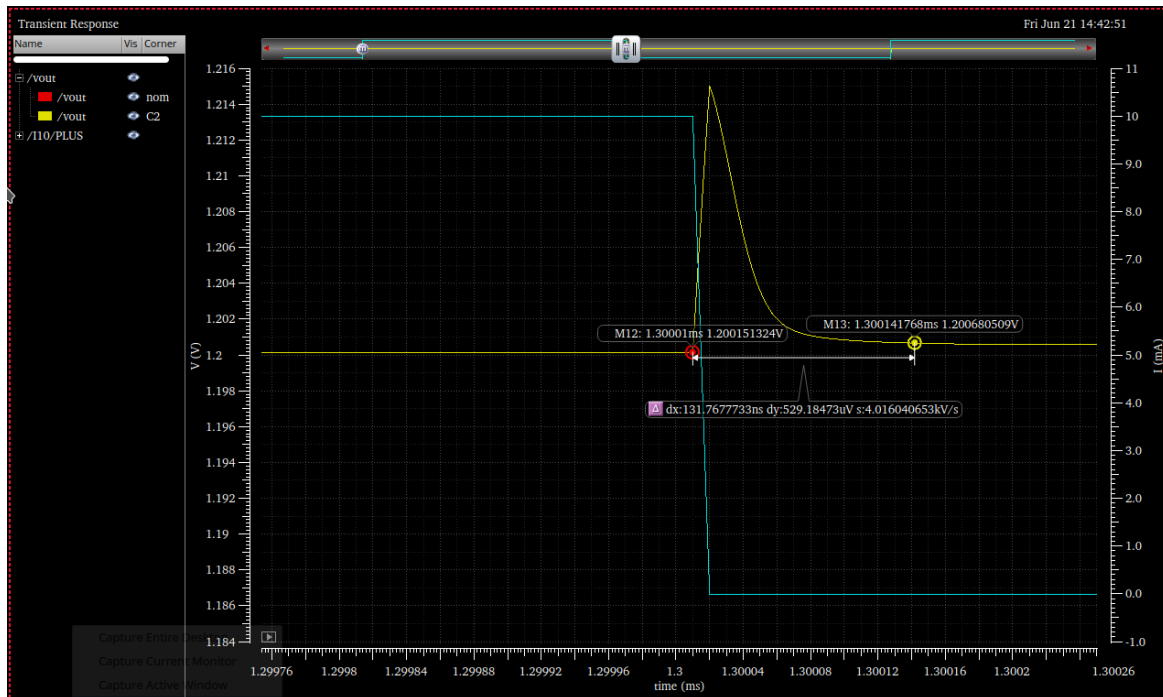


Figure 1.21: settling time falling FF

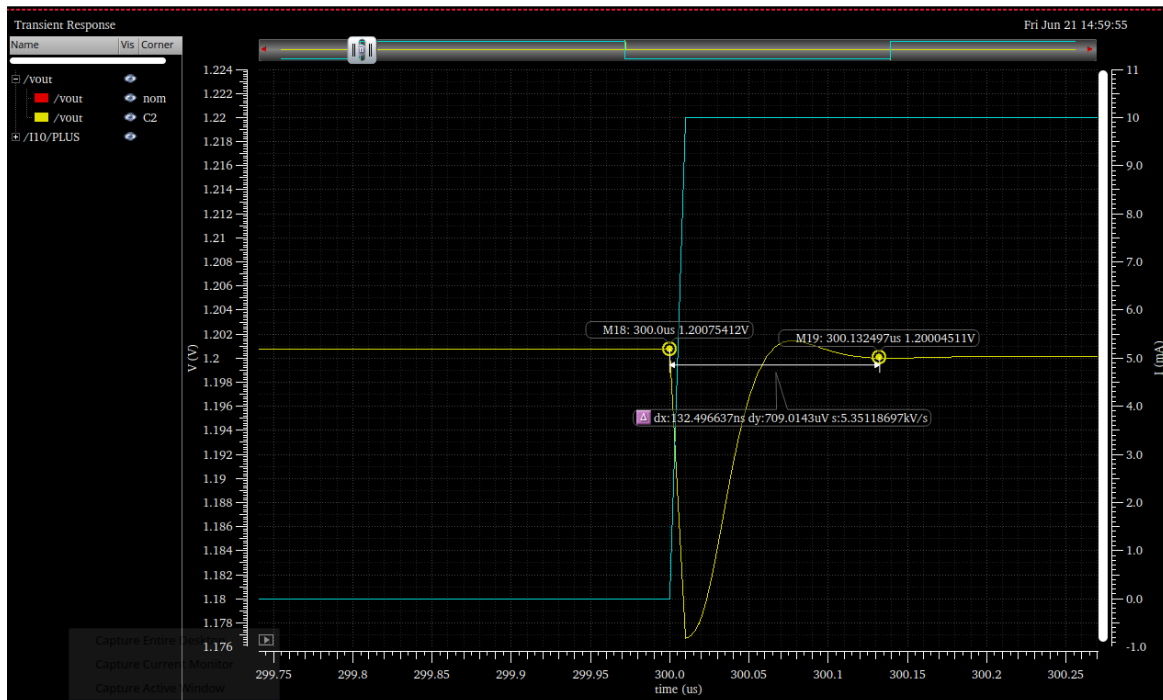


Figure 1.22: settling time rising SF

1.1 – POWER REGULATOR DESIGN QUESTIONS

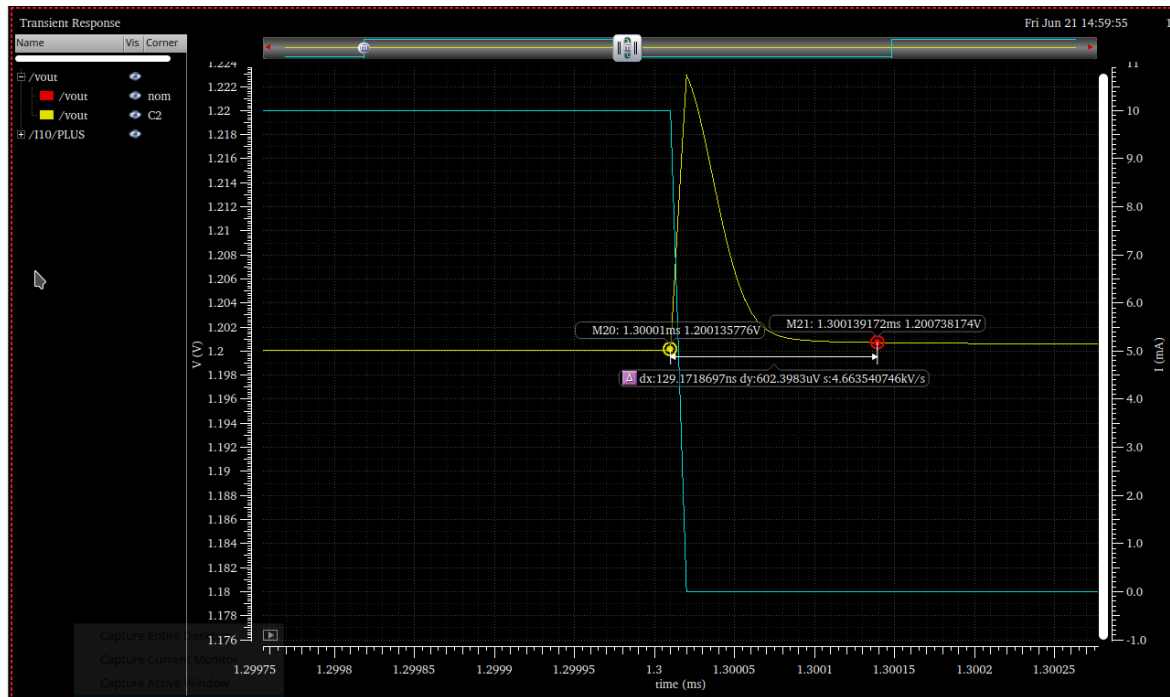


Figure 1.23: settling time falling SF

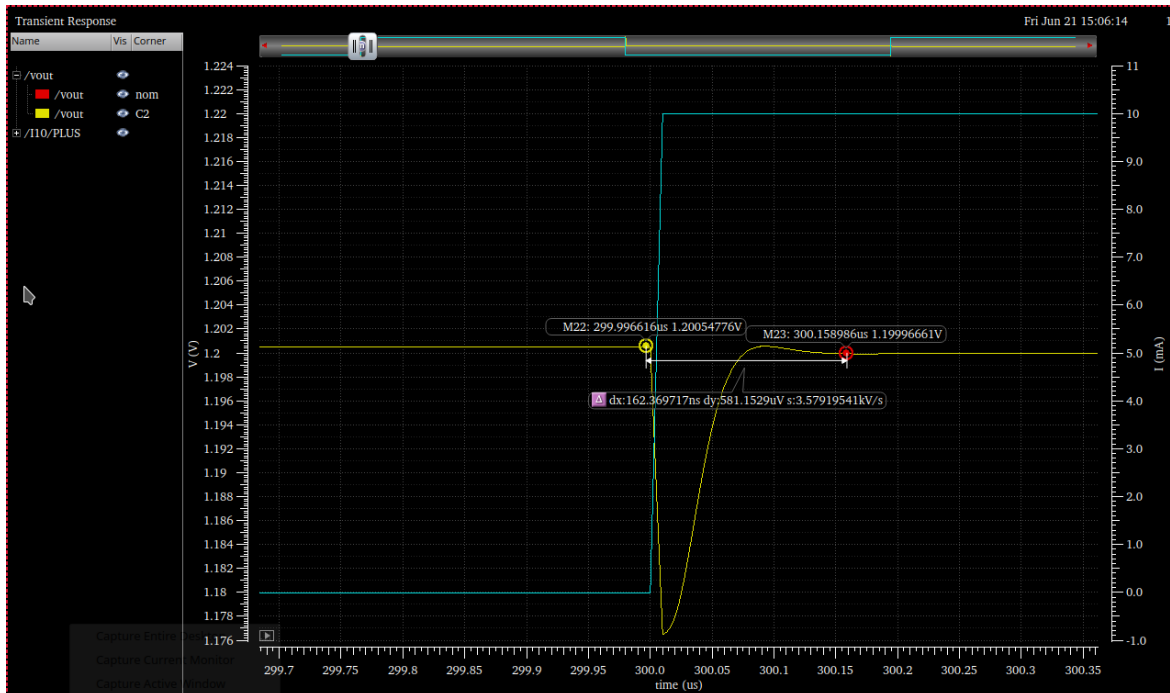


Figure 1.24: settling time rising SS

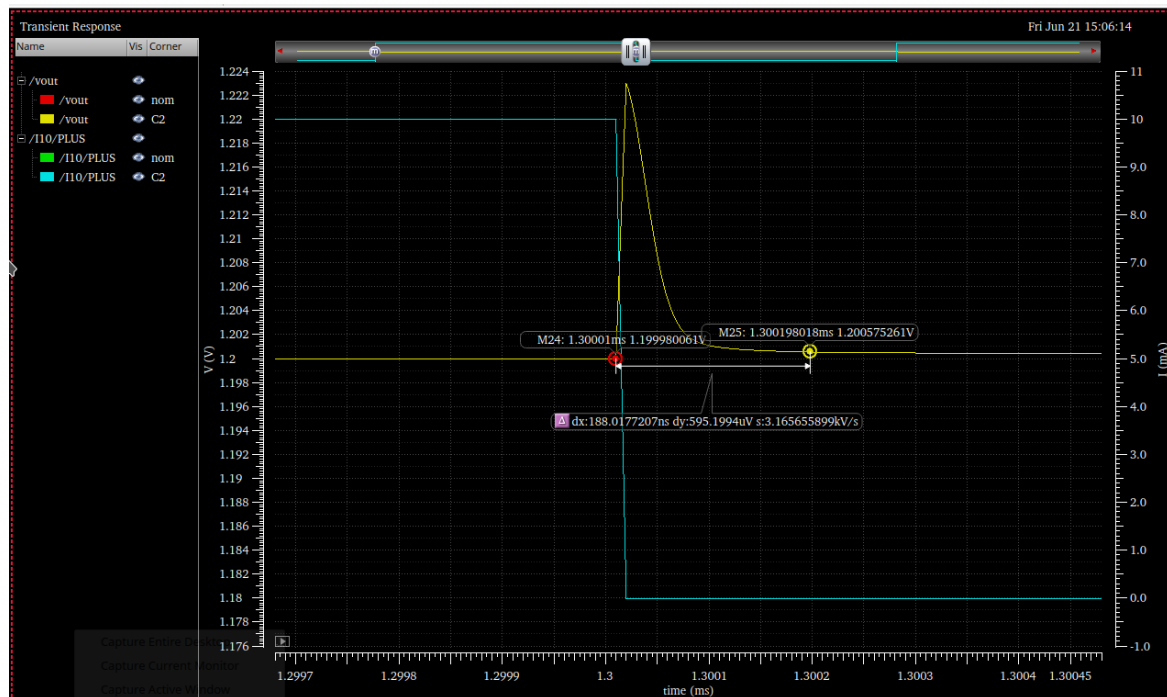


Figure 1.25: settling time falling SS

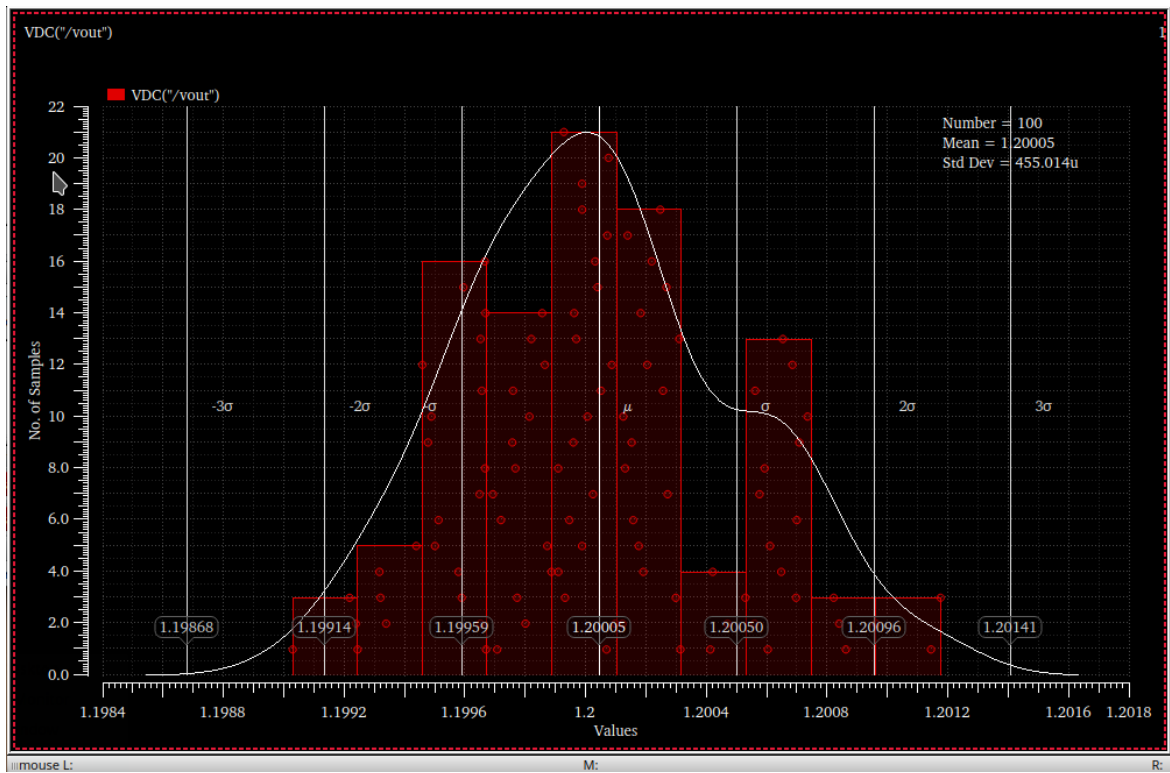
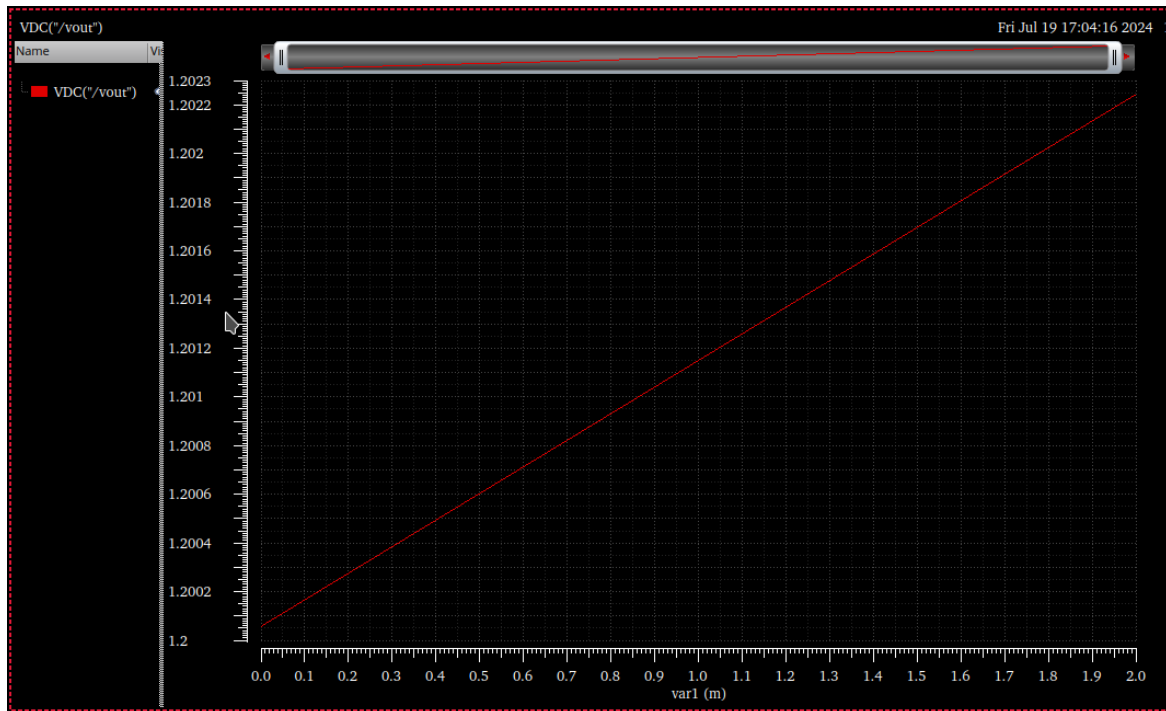


Figure 1.26: Vout monte carlo

Figure 1.27: Assume offset voltage V_{th}

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